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**Elements of
Optimal Control and Dynamic Programming**

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Introduction

Optimal control problems typically originate from the need to make decisions over time to optimize a specific objective. For example, in economics, firms may seek to maximize profit or minimize costs over time, leading to problems where the state of the system evolves according to economic laws and control inputs. In engineering, optimal control is used in systems such as robotics, aerospace, and process control, where the objective might be to minimize energy consumption, reduce time to completion, or improve system stability. These problems are generally formulated in terms of state variables that describe the system, control variables that can be manipulated, and an objective functional that quantifies the performance of the system under given controls.

We will both study deterministic and stochastic optimal control problems. The formers assume that the system dynamics and external influences are known with certainty. This framework is suitable for scenarios where the environment is highly predictable. However, many practical situations involve uncertainty, making the stochastic framework more appropriate. Stochastic control problems incorporate randomness into the system dynamics, leading to solutions that are robust to variations and uncertainties.

In Chapter 1 we first formalize the problem by defining the system dynamics through differential equations and specifying the control space. The value function is introduced to represent the optimal payoff. We then derive necessary conditions for optimality, such as the Hamilton-Jacobi-Bellman (HJB) equation, and prove verification theorems that establish these conditions as essentially sufficient for optimality. The Pontryagin Maximum Principle is then presented as an alternative method, offering a different perspective on deriving optimal controls by introducing the Hamiltonian and the costate variable. Finally, we discuss existence results for optimal controls under various conditions.

Chapter 2 mirrors the structure of Chapter 1 but extends the discussion to stochastic systems. The problem is first formalized with stochastic differential equations (SDEs) governing the system dynamics. We then derive the Stochastic Hamilton-Jacobi-Bellman (SHJB) equation as a necessary condition for optimality and prove stochastic verification theorems. The stochastic counterpart of the Pontryagin Maximum Principle, known as the Stochastic Maximum Principle, is introduced, involving backward stochastic differential equations (BSDEs) to characterize the optimal control. These mathematical objects are introduced and studied as well. Chapter 3 delves into the theory and application of viscosity solutions within the context of optimal control problems. In many instances, the value function arising from an optimal control problem may fail to be differentiable. This failure invalidates the derivation of the Hamilton-Jacobi equation and necessitates a weaker notion of solution to obtain a consistent and unique resolution to the dynamic programming equation for non-differentiable value functions. The viscosity solution is presented via a vanishing viscosity procedure, which allows the computation of the solution of a fully non-linear first-order partial differential equation (PDE) as the limiting solution of quasi-linear parabolic PDEs obtained via infinitesimal perturbations of second-order derivatives. Then, the formal definition of viscosity

solutions is provided, establishing conditions under which a function is considered a viscosity subsolution or supersolution. This definition is crucial for ensuring that the solution adheres to the dynamic programming principle even when the value function is not smooth. The chapter also presents an abstract dynamic programming framework, which generalizes the dynamic programming principle and facilitates the definition of viscosity solutions for the dynamic programming equation. The chapter further discusses the extension of the Verification Theorem and Pontryagin's principle to non-differentiable settings using viscosity solutions. Since the Hamiltonian may not be differentiable, the classical adjoint variable must be extended to accommodate non-smooth points by defining the sets of superdifferentials and subdifferentials.

Finally, chapter 4 addresses singular stochastic control problems, a special class of stochastic optimal control issues where control actions can cause instantaneous and discontinuous state changes. These problems are relevant in various fields such as finance (e.g., optimal trading strategies), manufacturing (e.g., optimal machine switching), and energy management (e.g., optimal resource extraction). The chapter begins by introducing the concept of singular stochastic control, emphasizing the need for controls that are not continuous functions of time but rather measures that can implement abrupt changes. The core of the chapter is dedicated to adapting and proving necessary conditions for optimality in this context, such as the Hamilton-Jacobi-Bellman (HJB) equation and verification theorems. The chapter also explores the use of viscosity solutions in the context of singular stochastic control.

The main bibliographical reference is the book 'Controlled Markov Processes and Viscosity Solutions' by Wendell H. Fleming and H.M. Soner[6]. It is a broad, technical, complete, and almost self-contained dissertation, which covers virtually all the topics in this thesis. Nevertheless, the Pontryagin's maximum principle is only briefly mentioned, thus we followed the lectures by Evans[3] for the deterministic setting, and the book by Huy  n Pham 'Continuous-time Stochastic Control and Optimization with Financial Applications'[9] for the stochastic counterpart. Finally, several other references were used to fill the technical gaps we encountered (e.g. lectures note by Varadhan[10], Bressan[1], or the book 'The Theory of Markov Processes' by Dynkin[2] among others cited in the bibliography).

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Chapter 1

Deterministic Optimal Control

This chapter begins by formalizing the optimal control problem, defining the system's state dynamics through differential equations and specifying the control space. The primary objective is to minimize or maximize a performance criterion represented by a value function, which depends on the state and control variables over a given time horizon. The chapter introduces the Hamilton-Jacobi-Bellman (HJB) equation as a necessary condition for optimality. This partial differential equation characterizes the value function and provides a foundation for deriving optimal controls. Verification theorems are then presented to establish that the HJB equation's conditions are essentially sufficient, ensuring that any function satisfying these conditions is indeed the optimal value function. In addition to the dynamic programming approach, the chapter explores the Pontryagin Maximum Principle. This principle offers an alternative method to derive optimal controls by introducing the Hamiltonian function and costate variables. It provides a set of necessary conditions that any optimal control must satisfy, offering a complementary perspective to the HJB equation. This chapter concludes with a discussion on existence theorems, providing rigorous proofs for the existence of optimal controls under various conditions.

1.1 Finite horizon

Let us consider a finite interval $I = [t, t_1] \subset \mathbb{R}$ as the operating time of a system. At each time $s \in I$, the system is described by $x(s) \in O \subseteq \mathbb{R}^n$ and controlled by $u(s) \in U \subseteq \mathbb{R}^m$ where U is called control space. The system's evolution is described by

$$\begin{cases} \dot{x}(s) = f(s, x(s), u(s)) & s \in I \\ x(t) = x, \end{cases} \quad (1.1)$$

for given $x \in O$ and suitable $f : \bar{Q} \times U \rightarrow \mathbb{R}^n$, where $Q = [t, t_1] \times O$. We impose $f \in C(\bar{Q} \times U)$ and the existence of $K_\rho > 0$ such that for all $\rho > 0$ then

$$|f(t, x, v) - f(t, y, v)| \leq K_\rho |x - y|, \quad (1.2)$$

for all $t \in I$, $x, y \in O$ and $v \in U$ such that $|v| \leq \rho$. Under this conditions the system (1.1) has a unique solution. Controls $u(\cdot)$ are assumed to be in the set $L^\infty([t, t_1]; U)$. We will soon specify more about the set of controls. We have described a control problem. The concept of optimality is related to a value function, specified by payoffs (or costs) associated to the system's states. Let $L \in C(\overline{Q} \times U)$ be the *running cost*, and $\Psi \in C(I \times O)$ the *terminal cost* defined as

$$\Psi(t, x) = \begin{cases} g(t, x) & \text{if } (t, x) \in [t, t_1) \times O \\ \psi(x) & \text{if } (t, x) \in \{t_1\} \times O. \end{cases} \quad (1.3)$$

We define the *payoff* J as

$$J(t, x; u) = \int_t^\tau L(s, x(s), u(s)) ds + \Psi(\tau, x(\tau)), \quad (1.4)$$

where τ is the exit time of $(s, x(s))$ from $\overline{Q}$, that is

$$\tau = \begin{cases} \inf\{s \in [t, t_1) \mid x(s) \notin \overline{O}\} & \text{if } \exists s \in [t, t_1) : x(s) \notin \overline{O} \\ t_1 & \text{if } x(s) \in \overline{O} \forall s \in [t, t_1). \end{cases} \quad (1.5)$$

A control $u^*(\cdot)$ is *optimal* if

$$J(t, x; u^*) \leq J(t, x; u) \quad \forall u \in L^\infty(I; U). \quad (1.6)$$

We have to impose a further condition on controls, the *switching condition*. Heuristically, we ask controls to be consistent as time passes, that is we can switch from one control to another at a given point in time without losing control's property. Let us assume that we have $u \in \mathcal{U}(t, x)$ and $u' \in \mathcal{U}(r, x(r))$ for $r \in [t, \tau]$. If we define

$$\tilde{u}(s) = \begin{cases} u(s) & s \in [t, r) \\ u'(s) & s \in [r, t_1], \end{cases} \quad (1.7)$$

then we impose

$$\tilde{u}_s \in \mathcal{U}(s, \tilde{x}(s)) \quad \forall s \in [t, \tilde{\tau}], \quad (1.8)$$

where $\tilde{x}$ is the solution to the control problem (1.1) with control $\tilde{u}$ and initial condition x , $\tilde{u}_s$ is the restriction of $\tilde{u}$ to $[s, t_1]$, and $\tilde{\tau}$ is the exit time of $(s, \tilde{x}(s))$ from $\overline{Q}$. This condition assures that admissible controls can be replaced as the time evolves and the resulting control is still admissible.

1.2 Dynamic programming principle

One way of tackling certain optimal control problems is via *dynamic programming*. The basic idea behind dynamic programming techniques is to subdivide a problem into smaller problems which are easier to solve. What does this mean in our context? We will be able to find an instantaneous condition on the problem, from

which we will retrieve an optimal control and the relative state variable. We need to define the *value function*:

$$V(t, x) = \inf_{u \in \mathcal{U}(t, x)} J(t, x; u) \quad (1.9)$$

for all $(t, x) \in \overline{Q}$. This value function V will be characterized by a partial differential equation (PDE) called Hamilton-Jacobi-Bellman equation, which will provide the necessary condition on the system to be optimized. We will see that this differential relation will also be some sort of sufficient condition on optimality.

We get rid of the instance in which $V(t, x) = -\infty$ by assuming Q to be compact, or L and Ψ to be bounded below. We aim at retrieving the argument which attains the infimum of (1.9). Prior to some exposure to dynamic programming a useful interpretation of the immersion of this optimal control problem into a dynamic programming one is to see the state of the system as the state of the variable and the control function as the decision function.

We start by stating and proving the following proposition, which provides us with an equivalent definition of the value function.

Proposition 1.2.1. *For any $(t, x) \in \overline{Q}$ and any $r \in I$ then*

$$V(t, x) = \inf_{u \in \mathcal{U}(t, x)} \left\{ \int_t^{r \wedge \tau} L(s, x(s), u(s)) ds + g(\tau, x(\tau)) \chi_{\tau < r} + V(r, x(r)) \chi_{r \leq \tau} \right\}. \quad (1.10)$$

Proof. Value function less than rhs. If $r > \tau$ then $\tau < t_1$ and $\Psi(r \wedge \tau, x(r \wedge \tau)) = g(\tau, x(\tau))$, thus (1.10) follows directly by definition. If $r \leq \tau$, let $\delta > 0$ then there exists $u^1 \in \mathcal{U}(r, x(r))$ such that

$$\int_r^{\tau^1} L(s, x^1(s), u^1(s)) ds + \Psi(\tau^1, x^1(\tau^1)) \leq V(r, x(r)) + \delta,$$

where x^1 is the state function corresponding to u^1 with initial condition $(r, x(r))$ and τ^1 the first exit from $\overline{Q}$ of $(s, x^1(s))$. By defining $\tilde{u}$ as for the switching condition (1.7) we have $\tau^1 = \tilde{\tau}$, because $\tau \geq r$, and then $\tilde{u}$ is u^1 . Then

$$\begin{aligned} V(t, x) &\leq V(t, x; \tilde{u}) \\ &= \int_t^{\tilde{\tau}} L(s, \tilde{x}(s), \tilde{u}(s)) ds + \Psi(\tilde{\tau}, \tilde{x}(\tilde{\tau})) \\ &= \int_t^r L(s, x(s), u(s)) ds + \int_r^{\tau^1} L(s, x^1(s), u^1(s)) ds + \Psi(\tau^1, x^1(\tau^1)) \\ &\leq \int_t^r L(s, x(s), u(s)) ds + V(r, x(r)) + \delta. \end{aligned}$$

Since δ is arbitrary the first inequality is proved.

Value function is bigger than rhs. Let $\delta > 0$ and $u \in \mathcal{U}(t, x)$ such that

$$\int_r^\tau L(s, x(s), u(s)) ds + \Psi(\tau, x(\tau)) \leq V(t, x) + \delta,$$

then

$$\begin{aligned} V(t, x) &\geq \int_r^\tau L(s, x(s), u(s)) ds + \Psi(\tau, x(\tau)) - \delta \\ &= \int_t^{r \wedge \tau} L(s, x(s), u(s)) ds + \int_{r \wedge \tau}^\tau L(s, x(s), u(s)) ds + \Psi(\tau, x(\tau)) - \delta \\ &= \int_t^{r \wedge \tau} L(s, x(s), u(s)) ds + J(r, x(r))\chi_{r \leq \tau} + g(\tau, x(\tau))\chi_{\tau < r} - \delta \\ &= \int_t^{r \wedge \tau} L(s, x(s), u(s)) ds + V(r, x(r))\chi_{r \leq \tau} + g(\tau, x(\tau))\chi_{\tau < r} - \delta, \end{aligned}$$

since δ is arbitrary we proved the proposition. $\square$

In the proof we used the concept of δ -optimal control, that is a control function $u \in \mathcal{U}(r, x(r))$ such that

$$\int_r^{\tau^1} L(s, x^1(s), u^1(s)) ds + \Psi(\tau^1, x^1(\tau^1)) \leq V(r, x(r)) + \delta.$$

This new representation allows us to find the so-called *dynamic programming equation*. We have to impose that the value function is continuously differentiable, although this is not always the case. If differentiability fails, the notion of viscosity solution is needed; we will delve into it later.

Let us first impose boundary conditions of the value function. Clearly if $t = t_1$ then

$$V(t_1, x) = \psi(x) \quad \forall x \in \overline{O}. \quad (1.11)$$

If $(t, x) \in [t_0, t_1) \times \partial O$ then the value function is g :

$$V(t, x) = g(t, x). \quad (1.12)$$

Before stating the fundamental theorem which gives sufficient conditions for a solution to the optimal problem, we follow a heuristic reasoning which will help our intuition. Under the hypothesis of continuous differentiability of the value function let us rewrite the dynamic programming principle as:

$$\begin{aligned} \inf_{u \in \mathcal{U}} \left\{ \frac{1}{h} \int_t^{(t+h) \wedge \tau} L(s, x(s), u(s)) ds + \frac{1}{h} g(\tau, x(\tau))\chi_{\tau < t+h} \right. \\ \left. + \frac{1}{h} [V(t+h, x(t+h))\chi_{\tau \geq t+h} - V(t, x)] \right\} = 0. \end{aligned} \quad (1.13)$$

Then, if we formally let $h \rightarrow 0$, we get

$$\inf_{u \in \mathcal{U}} \{L(t, x(t), u(t)) + \partial_t V(t, x(t)) + D_x V(t, x(t)) \cdot f(t, x(t), u(t))\} = 0,$$

which can be rewritten as

$$-\frac{\partial}{\partial t} V(t, x) + H(t, x, D_x V(t, x)) = 0, \quad (1.14)$$

where for $(t, x, p) \in \overline{Q} \times \mathbb{R}^n$ the Hamiltonian is defined as

$$H(t, x, p) = \sup_{v \in \mathbb{R}^n} \{-p \cdot f(t, x, v) - L(t, x, v)\}. \quad (1.15)$$

Equation (1.14) turns out to be the key sufficient condition for the value function to be optimal.

Theorem 1.2.2 (Verification Theorem). *Let $W \in C^1(\overline{Q})$ satisfy (1.14) and the boundary conditions (1.11) and (1.12) then*

$$W(t, x) \leq V(t, x) \quad \forall (t, x) \in \overline{Q}.$$

Moreover, there exists $u^* \in \mathcal{U}$ such that:

$$\begin{cases} L(s, x^*(s), u^*(s)) + f(s, x^*, u^*(s)) \cdot D_x W(s, x^*(s)) \\ \quad \quad \quad = -H(s, x^*(s), D_x W(s, x^*(s))) & \text{a.s. for } s \in [t, \tau^*] \\ W(\tau^*, x^*(\tau^*)) = g(\tau^*, x^*(\tau^*)) & \text{if } \tau^* < t_1, \end{cases} \quad (1.16)$$

if and only if u^* is optimal and $W = V$.

Proof. Let $u \in \mathcal{U}$, then:

$$\begin{aligned} \Psi(\tau, x(\tau)) &= W(\tau, x(\tau)) \\ &= W(t, x(t)) + \int_t^\tau \frac{d}{ds} W(s, x(s)) ds \\ &= W(t, x(t)) + \int_t^\tau \left(\frac{\partial}{\partial t} W(s, x(s)) + \dot{x}(s) \cdot D_x W(s, x(s)) \right) ds \\ &= W(t, x(t)) + \int_t^\tau \left(\frac{\partial}{\partial t} W(s, x(s)) + f(s, x(s), u(s)) \cdot D_x W(s, x(s)) \right) ds \\ &\stackrel{(*)}{\geq} W(t, x(t)) - \int_t^\tau L(s, x(s), u(s)) ds \end{aligned}$$

then:

$$W(t, x(t)) \leq J(t, x; u),$$

therefore, by taking the infimum over $\mathcal{U}$ and recalling $x(t) = x$ we get

$$W(t, x) \leq V(t, x).$$

If furthermore u^* satisfies (1.16) then the inequality $\stackrel{\circ}{\geq}$ is an equality, and thus

$$W(t, x) = J(t, x; u^*),$$

which implies that u^* is optimal and $W(t, x) = J(t, x; u^*) = V(t, x)$.

The converse is immediate by definition of H in (1.15) for the first part of (1.16) and $W = V$ for the second part. □

Theorem 1.2.2 is an important tool in determining the explicit form of an optimal control. Indeed, condition (1.16) can be restated as

$$u^*(s) \in \arg \min_{v \in U} \{f(s, x^*(s), v) \cdot D_x W(s, x^*(s)) + L(s, x^*(s), v)\}, \quad (1.17)$$

for almost all $s \in [t, t_1]$.

We can express the optimality condition on u (1.17) in a differential inclusion form.

Corollary 1.2.3. *A control u^* is optimal if the corresponding state function x^* satisfies*

$$x^* \in \{f(t, x, v) \mid v \in v^*(t, x)\}. \quad (1.18)$$

1.3 Pontryagin's principle and dynamic programming

In the previous section we tackled the optimal control problem via dynamic programming. As mentioned earlier this approach is of wide applicability and provides an implicit characterization of an optimal control. We now present another technique: the Pontryagin's principle. As before, it will give rise to necessary conditions on a control function to be optimal, but they'll come from a completely different perspective. We now present Pontryagin's principle in its full generality, and then we will see how it is connected with the dynamic programming approach.

1.3.1 Pontryagin's principle

Pontryagin gives us an elegant and unintuitive way of solving problems of the kind

$$\begin{cases} \dot{x}(s) = f(s, x(s), u(s)) & s \in [t, t_1] \\ x(t) = x, \end{cases} \quad (1.19)$$

where $u(\cdot)$ is bounded measurable into $U \subset \mathbb{R}^m$ and $O = \mathbb{R}^n$. Having defined the functional $J(t, x; u)$ as usual we define the *control state Hamiltonian* as follows.

Definition 1.3.1. *The control state Hamiltonian of system (1.19) is*

$$H(s, x, u, p) = -p \cdot f(s, x, u) - L(s, x, u), \quad (1.20)$$

for all $s \in [t, t_1]$, $x, p \in O$, $u \in U$.

The variable p is called *costate* of the system. Pontryagin's principle gives us information about the costate under an optimal trajectory, which in turn will characterize the optimal control.

Theorem 1.3.1. *Let u^* be an optimal control and x^* its corresponding trajectory. Then there exists a function $p^* : [t, t_1] \rightarrow O$ such that:*

$$\dot{x}^*(s) = D_p H(s, x^*(s), u^*(s), p^*(s)) \quad (1.21)$$

$$\dot{p}^*(s) = -D_x H(s, x^*(s), u^*(s), p^*(s)) \quad (1.22)$$

And also:

$$H(s, x^*(s), u^*(s), p^*(s)) = \sup_{v \in U} H(s, x^*(s), v, p^*(s)) \quad (1.23)$$

With:

$$p^*(t_1) = D\psi(x^*(t_1)) \quad (1.24)$$

Thanks to this result we can determine an optimal control via the costate. By solving equation (1.22) arisen in Theorem 1.3.1 we can obtain the explicit form of p^* , and then retrieve u^* from the maximization principle (1.23).

1.3.2 Dynamic programming interplay

As we just saw, Pontryagin's principle offers us a technique for finding an optimal control, although the necessary condition is quite involved. The control state Hamiltonian is crucial in this approach, it reminds us of the Hamiltonian defined in (1.15). The similarity is also fortified by the maximization principle (1.23). We shall prove that this similarity unveils the direct link between Pontryagin's principle and the dynamic programming equation. We will use the notion of differentiability of the value function.

Definition 1.3.2. *V is differentiable in (t, x) if there exists $V_t(t, x), V_x(t, x) \in \mathbb{R}$ such that*

$$\lim_{(h,k) \rightarrow (0,0)} \frac{1}{|h| + |k|} |V(t+h, x+k) - V(t, x) - V_t(t, x) \cdot h - V_x(t, x) \cdot k| = 0. \quad (1.25)$$

Differentiability is somewhat a strong hypothesis, but it allows us to prove the following proposition. We must say that differentiability may easily fail in application, in such instances the notion of a weaker solution is needed, namely a viscosity solution.

Theorem 1.3.2. *Let V be differentiable in $(t, x) \in Q$ and u^* an optimal control such that $u^* \xrightarrow{s \rightarrow t} v$, then:*

$$V_t(t, x) + L(t, x, v) + f(t, x, v) \cdot D_x V(t, x) = 0 \quad (1.26)$$

Proof. Let $h > 0$ s.t. $t + h < \tau$, then by (1.2.1) we have

$$V(t, x) = \int_t^{t+h} L(s, x(s), u^*(s)) ds + V(t + h, x(t + h)),$$

because of differentiability we have

$$\lim_{h \rightarrow 0} \frac{1}{|h|} |V(t + h, x(t + h)) - V(t, x(t))| = V_t(t, x) + f(t, x, v) \cdot D_x V(t, x).$$

Thus we get

$$\begin{aligned} L(t, x, v) &= \lim_{h \rightarrow 0} \frac{1}{|h|} \int_t^{t+h} L(s, x(s), u(s)) ds = - \lim_{h \rightarrow 0} \frac{1}{|h|} |V(t + h, x(t + h)) - V(t, x(t))| \\ &= -V_t(t, x) - f(t, x, v) \cdot D_x V(t, x). \end{aligned}$$

□

Furthermore, we impose existence and continuity of all derivatives of f, L, g, ψ . The next theorem demonstrates that the costate in the Pontryagin Maximum Principle is in fact the gradient in x of the value function $v(\cdot)$, taken along an optimal trajectory.

Theorem 1.3.3. *Let u^* be an optimal right-continuous control and x^* its corresponding trajectory. Assume that the value function V is differentiable at $(s, x^*(s))$ for $s \in [t, t_1]$. Let*

$$p(s) = D_x V(s, x^*(s)). \quad (1.27)$$

Then $p(s)$ satisfies (1.22), (1.23) and (1.24).

Proof. The transversality condition (1.24) is straightforward from the definition of the value function, which implies also the maximality condition (1.23). We need to prove the "lagrangian multiplier condition" (1.22). Let us drop all $*$ and rewrite this differential equation as

$$\frac{d}{dt} p_j(s) = - \sum_{i=1}^n \frac{\partial}{\partial x_j} f_i(s, x(s), u(s)) p_i(s) - \frac{\partial}{\partial x_j} L(s, x(s), u(s)). \quad (1.28)$$

This system admits solution $\bar{p}$ such that $\bar{p}(s) = D_x V(s, x(s))$. We show that $\bar{p}(s) = p(s)$ for all $s \in [t, t_1]$. Let u_s be the restriction of u to $[s, t_1]$, which is admissible by assumption. We have

$$V(s, y) \leq J(s, y; u_s) \quad \forall y \in \mathbb{R}^n,$$

then, because u is optimal, $y \mapsto J(s, y; u_s) - V(s, y)$ has its global minimum in $y = x(s)$, which implies, by differentiability that

$$D_x V(s, x(s)) = D_x J(s, x(s); u_s). \quad (1.29)$$

Then we prove that $\bar{p}(s) = D_x J(s, x(s); u_s)$. We denote $x(r)$ the solution starting at $x(s)$ at time r . Because $L \in C^1$ then for all $i = 1, \dots, n$ we have

$$\frac{\partial}{\partial x_i} J(s, x(s), u) = \sum_{j=1}^n \int_s^{t_1} \left(L_{x_j}(r, x(r), u(r)) \frac{\partial x_j(r)}{\partial x_i} \right) dr + \psi_{x_j}(x(t_1)) \frac{\partial x_j(t_1)}{\partial x_i},$$

thus

$$\begin{aligned} \bar{p}_i(s) &= \sum_{j=1}^n \frac{\partial x_j(s)}{\partial x_i} \bar{p}_j(s) \\ &= \sum_{j=1}^n \frac{\partial x_j(t_1)}{\partial x_i} \bar{p}_j(t_1) - \int_s^{t_1} \frac{d}{dr} \left(\sum_{j=1}^n \frac{\partial x_j(r)}{\partial x_i} \bar{p}_j(r) \right) dr \\ &= \sum_{j=1}^n \frac{\partial x_j(t_1)}{\partial x_i} \psi(x(t_1)) - \sum_{j=1}^n \int_s^{t_1} \frac{d}{dr} \left(\frac{\partial x_j(r)}{\partial x_i} \right) \bar{p}_j(r) + \frac{\partial x_j(r)}{\partial x_i} \frac{d}{dr} (\bar{p}_j(r)) dr. \end{aligned}$$

But, under the integral:

$$\begin{aligned} &\int_s^{t_1} \frac{d}{dr} \sum_{l=1}^n \frac{\partial x_l(r)}{\partial x_i} \left(\frac{\partial}{\partial x_j} f_l(r, x(r), u(r)) \bar{p}_j(r) \right. \\ &\quad \left. - \sum_{l=1}^n \frac{\partial x_l(r)}{\partial x_i} \frac{\partial}{\partial x_j} f_l(r, x(r), u(r)) \bar{p}_j(r) - \frac{\partial}{\partial x_j} L(r, x(r), u(r)) \right) dr \end{aligned}$$

Finally,

$$\begin{aligned} \bar{p}_i(s) &= \sum_{j=1}^n \left(\frac{\partial x_j(t_1)}{\partial x_i} \psi(x(t_1)) + \int_s^{t_1} \frac{\partial}{\partial x_j} L(r, x(r), u(r)) \frac{\partial x_j(r)}{\partial x_i} dr \right) \\ &= \frac{\partial}{\partial x_i} J(s, x(s); u_r). \end{aligned}$$

□

1.4 Existence

We now prove an existence theorem for optimal controls. We study the fixed time interval case with $O = \mathbb{R}^n$ and the function f linear in v . Furthermore, we impose convexity of L in v . Under these assumptions a classical variational argument proves the optimal control existence.

Theorem 1.4.1. *Let U compact and convex, $f_1, f_2 \in C^1(\overline{Q} \times U)$ such that $f(t, x, v) = f_1(t, x) + f_2(t, x)v$ and $\partial_x f_1, \partial_x f_2, f_2$ bounded. Let also $L \in C^1(\overline{Q} \times U)$ and $L(t, x, \cdot)$ be convex for all $(t, x) \in \overline{Q}$ and the terminal cost $\psi \in C(\mathbb{R}^n)$. Then there exist an optimal control $u^*(\cdot)$.*

Proof. Let u_n a minimizing sequence such that

$$\lim_{n \rightarrow +\infty} J(t, x; u_n) = V(t, x). \quad (1.30)$$

Let $x_n(\cdot)$ be the solutions to (1.1) with $u = u_n$. If we show both sequence to converge respectively (weakly) to u^* and uniformly x^* (along subsequences) such that the latter is again the solution to (1.1) with $u = u^*$, then

$$\begin{aligned} J(t, x; u_n) &= \int_t^{t_1} L(s, x^*(s), u_n(s)) ds + \int_t^{t_1} L(s, x_n(s), u_n(s)) \\ &\quad - L(s, x^*(s), u_n(s)) ds + \phi(x_n(t_1)) \end{aligned}$$

implies

$$\liminf_{n \rightarrow +\infty} \int_t^{t_1} L(s, x_n(s), u_n(s)) - L(s, x^*(s), u_n(s)) ds = 0$$

and $\psi(x_n(t_1)) \xrightarrow{n \rightarrow +\infty} \psi(x^*(t_1))$. But then

$$\liminf_{n \rightarrow +\infty} J(t, x; u_n) = \liminf_{n \rightarrow +\infty} \int_t^{t_1} L(s, x^*(s), u_n(s)) ds \geq \int_t^{t_1} L(s, x^*(s), u^*(s)) ds, \quad (1.31)$$

because L is convex in u ^[1]. Therefore

$$V(t, x) \leq J(t, x; u^*) \leq \liminf_{n \rightarrow +\infty} J(t, x; u_n) = V(t, x).$$

We need to prove the convergence of x_n and u_n . Because U is compact and *convex* then $L^\infty([t, t_1]; U)$ is weakly sequentially compact. For what concerns x_n we use Ascoli-Arzelà's theorem to show that it admits a uniformly convergent subsequence. Being uniformly limited comes from:

^[1]Explained in remark 1.4

$$\begin{aligned}
 |x_n(s)| &\leq |x_n(s) - x| + |x| = |x| + \int_t^s \frac{d}{dr} |x_n(r)| dr \\
 &= |x| + \int_t^s |f_1(r, x_n(r)) + f_2(r, x_n(r)) \cdot u_n(r)| dr \\
 &\leq |x| + \int_t^s \|\partial_x f_1\|_\infty |x_n(r)| + \|f_2\|_\infty |u_n| dr \\
 &\leq C + K \left(\int_t^s |x_n(r)| dr \right)
 \end{aligned}$$

then by Gronwall's lemma x_n is uniformly limited. Equicontinuity comes from the uniform boundedness of the derivative $\dot{x}_n(s)$. Therefore, we know that there exist the weak limit u^* and the uniform limit x^* . The latter is the solution of (1.1) with $u = u^*$. Indeed:

$$\begin{aligned}
 x_n(s) &= x + \int_t^s \frac{d}{dr} x_n(r) dr \\
 &= x + \underbrace{\int_t^s f_1(r, x_n(r)) + f_2(r, x_n(r)) u^*(r) dr}_{A_n} \\
 &\quad + \underbrace{\int_t^s [f_2(r, x_n(r)) - f_2(r, x^*(r))] [u_n(r) - u^*(r)] dr}_{B_n} \\
 &\quad + \underbrace{\int_t^s f_2(r, x^*(r)) [u_n(r) - u^*(r)] dr}_{C_n}
 \end{aligned}$$

thus letting $n \rightarrow +\infty$ we get B_n (by weak convergence and boundedness of f_2) and C_n (by weak convergence) going to 0 while we obtain

$$A_n \xrightarrow{n \rightarrow +\infty} \int_t^s f_1(r, x^*(r)) + f_2(r, x^*(r)) u^*(r) dr,$$

and therefore the thesis. $\square$

Remark. In the proof we asserted inequality (1.31) by convexity of the running cost. Indeed, by convexity and being C^1 :

$$L(s, x(s), u_n(s)) \geq L(s, x(s), u^*(s)) + [u_n(s) - u^*(s)] L_u(s, x(s), u^*(s))$$

Then by integrating and taking $\liminf_{n \rightarrow +\infty}$ we get the inequality (using weak convergence of u_n).

An existence result can also be proved in the context of $O \neq \mathbb{R}^n$, where the cost function to be minimized has the form

$$J(t, x; u) = \int_t^\tau L(s, x(s), u(s)) ds + \Psi(\tau, x(\tau)), \quad (1.32)$$

with dynamical system described by $f(s, x, u) = u$. Therefore, the value function is

$$V(t, x) = \inf_{u \in \mathcal{U}} \left\{ \int_t^\tau L(s, x(s), \dot{x}(s)) ds + \Psi(\tau, x(\tau)) \right\}. \quad (1.33)$$

Under technical assumptions an optimal control exists, and the proof follows the same procedure as before: we take a minimizing sequence, show its (weak) convergence and its corresponding state evolution (uniform) convergence, then we show optimality, slightly modifying the proof because of convergence issues. We could consider more general $f(\cdot, \cdot, \cdot)$ and still get existence, indeed as [p]rove the key conditions on f is for the

$$f(t, x, U) = \{f(t, x, v) : v \in U\}$$

to be convex. Indeed, the Following holds true.

Theorem 1.4.2. *Let U compact and f Lipschitz with respect to the state variable and such that $f(t, x, U)$ is convex for all (t, x) , then there exists an optimal control for all (t, x) .*

1.5 Proof of Pontryagin's principle

We will show the maximum Pontryagin's principle in the simple context of no running cost. A clever reconstruction of the non-zero running cost problem as a zero running cost one will enlarge the thesis to the broader setting. The first concept we will need is the variation of a control.

Definition 1.5.1. *Given $u \in \mathcal{U}$. For $\epsilon, r > 0$ such that $0 < r - \epsilon < r$ and $a \in U$ we define the simple variation $u_\epsilon \in \mathcal{U}$ such that*

$$u_\epsilon(t) = \begin{cases} a & s \in (r - \epsilon, r) \\ u(s) & s \notin (r - \epsilon, r). \end{cases} \quad (1.34)$$

By defining the matrix $A : [0, +\infty) \rightarrow \mathbb{R}^{n \times n} : s \mapsto D_x f(s, x(s), u(s))$ we can state the following lemma.

Lemma 1.5.1. *Let x_ϵ be solution of*

$$\begin{cases} \dot{x}_\epsilon(s) = f(s, x_\epsilon(s), u_\epsilon(s)) & s \in [t, t_1] \\ x_\epsilon(t) = x, \end{cases} \quad (1.35)$$

then

$$x_\epsilon(s) = x(s) + \epsilon y(s) + o(\epsilon) \quad \epsilon \rightarrow 0, \quad (1.36)$$

where $y \equiv 0$ on $[t, r]$ and

$$\begin{cases} \dot{y}(s) = A(s)y(s) & s \in [r, t_1] \\ y(r) = y^r, \end{cases} \quad (1.37)$$

with $y^r = f(x(r), a) - f(x(r), u(r))$.

Proof. Let us divide the proof in three cases.

- $s \in [t, r - \epsilon]$: then $y(t) = 0$ and $u_\epsilon(t) = u(t)$, therefore:

$$x_\epsilon(t) = x(t) = x(t) + \epsilon y(t) + o(\epsilon)$$

- $s \in (r - \epsilon, r)$: then we have:

$$x_\epsilon(s) - x(s) = \int_{r-\epsilon}^s f(w, x_\epsilon(w), u_\epsilon(w)) - f(w, x(w), u(w)) dw + o(\epsilon)$$

Which is a little o of ϵ (because f is continuous).

- $s \in [r, t_1]$: from before if $s = r$ then:

$$\begin{aligned} x_\epsilon(r) - x(r) &= \int_{r-\epsilon}^r f(w, x_\epsilon(w), u_\epsilon(w)) - f(w, x(w), u(w)) dw + o(\epsilon) \\ &= \lim_{w \rightarrow s} [f(w, x_\epsilon(w), u_\epsilon(w)) - f(w, x(w), u(w))] \epsilon + o(\epsilon) \\ &= y^s \epsilon + o(\epsilon) \end{aligned}$$

If $s > r$ then $x(\cdot)$ and $x_\epsilon(\cdot)$ solve the same ODE with initial condition at s given by^[2]

$$x_\epsilon(s) = x(s) + \epsilon(f(x(s), a) - f(x(s), u(s))) + o(\epsilon),$$

thus, it can be shown^[3] that this necessarily implies

$$x_\epsilon(s) = x(s) + \epsilon y(s) + o(\epsilon).$$

□

^[2]via first order approximation.

^[3]Evans

Let us now prove Pontryagin's principle with no running cost. The payoff functional is

$$J(t, x; u) = \psi(x(t_1)), \quad (1.38)$$

and therefore the Hamiltonian is

$$H(s, x, u, p) = -f(s, x, u) \cdot p. \quad (1.39)$$

Theorem 1.5.2. *There exists a function $p^* : [t, t_1] \rightarrow \mathbb{R}^n$ such that*

$$\dot{p}^*(s) = -D_x H(s, x^*(s), u^*(s), p^*(s)) \quad s \in [t, t_1], \quad (1.40)$$

together with the maximization problem

$$H(s, x^*(s), u^*(s), p^*(s)) = \sup_{v \in U} H(s, x^*(s), v, p^*(s)), \quad (1.41)$$

and the transversality condition

$$p^*(t_1) = D\psi(x(t_1)). \quad (1.42)$$

Proof. Let us drop all the *. Let p be the unique solution of:

$$\begin{cases} \dot{p}(s) = -A'(s) \cdot p(s) & s \in [t, t_1] \\ p(t_1) = D\psi(x(t_1)) \end{cases} \quad (1.43)$$

It exists and is unique because the latter is a linear differential equation with integrable coefficient. We already satisfy the transversality condition and the adjoint dynamics. We prove the maximization principle. Let $a \in U$. We define the variation u_ϵ for $\epsilon, r \in (t, t_1)$ as before. Since $\epsilon \mapsto J(t, x; u_\epsilon)$ for $\epsilon \in [0, 1]$ has a maximum in $\epsilon = 0$ we have:

$$\frac{d}{d\epsilon} J(t, x; u_\epsilon) \Big|_{\epsilon=0} \leq 0.$$

Computing the derivative, using (1.34) we have

$$\begin{aligned} \frac{d}{d\epsilon} J(t, x; u_\epsilon) \Big|_{\epsilon=0} &= \frac{d}{d\epsilon} \psi(x_\epsilon(t_1)) \Big|_{\epsilon=0} \\ &= \frac{d}{d\epsilon} \psi(x(t_1) + \epsilon y(t_1) + o(\epsilon)) = D\psi(x(t_1)) \cdot y(t_1) \\ &= p(t_1) \cdot y(t_1) = p(r) \cdot [f(r, x(r), a) - f(r, x(r), u(r))], \end{aligned}$$

where the last equality comes from

$$\begin{aligned} \frac{d}{ds} (p(s) \cdot y(s)) &= \dot{p}(s) \cdot y(s) + p(s) \cdot \dot{y}(s) \\ &= -A'(s) \cdot p(s) \cdot y(s) + p(s) \cdot A(s) \cdot y(s) = 0. \end{aligned}$$

Therefore, by plugging into (1.5.2) we get

$$0 \geq p(r) \cdot [f(r, x(r), a) - f(r, x(r), u(r))],$$

which implies

$$H(r, x(r), a, p(r)) = f(r, x(r), a) \cdot p(r) \leq f(r, x(r), u(r)) \cdot p(r) = H(r, x(r), u(r), p(r)).$$

□

Given Pontryagin's principle for no running cost problems we can extend the result to the general case

$$J(t, x; u) = \int_t^{t_1} L(s, x(s), u(s)) ds + \psi(x(t_1))$$

where the Hamiltonian is

$$H(s, x, u, p) = f(s, x, u) \cdot p + L(s, x, u).$$

Indeed, theorem 1.5.2 holds also under these conditions. We rewrite the problem as it has no running cost and then apply the theorem. Let us define x^{n+1} as

$$x^{n+1}(s) = \int_t^s L(w, x(w), u(w)) dw.$$

Then, by defining $\bar{f}, \bar{g}, \bar{x}, \bar{x}(s)$ as

$$\bar{f}(s, x, u) = \begin{bmatrix} f(s, x, u) \\ L(s, x, u) \end{bmatrix}, \bar{x} = \begin{bmatrix} x \\ 0 \end{bmatrix}, \bar{x}(s) = \begin{bmatrix} x(s) \\ x^{n+1}(s) \end{bmatrix}, \bar{g}(\bar{x}(t_1)) = g(x(t_1)) + x^{n+1}(t_1),$$

the problem has no running cost, thus we can apply the theorem and, noticing that $p^{n+1} \equiv 1$, we get the thesis.

Chapter 2

Stochastic Optimal Control

This chapter provides an introduction to stochastic optimal control, paralleling the deterministic framework. We begin by formalizing the optimal control problem in stochastic settings, where system dynamics are influenced by random disturbances. The state dynamics are defined using stochastic differential equations (SDEs), and the control space is specified. The goal is to optimize a performance criterion, represented by a value function dependent on state and control variables over time. The chapter introduces the Stochastic Hamilton-Jacobi-Bellman (SHJB) equation as a necessary condition for optimality. This equation characterizes the value function in the stochastic context, providing a basis for deriving optimal controls. Verification theorems are presented to show that the conditions derived from the SHJB equation are essentially sufficient, ensuring that any function satisfying these conditions is the optimal value function. Following the dynamic programming approach, the chapter explores the Stochastic Maximum Principle. This principle extends Pontryagin's Maximum Principle to stochastic systems, introducing the Hamiltonian function and costate variables in the stochastic setting. It provides necessary conditions for optimality, involving backward stochastic differential equations (BSDEs) to capture the stochastic nature of the problem. BSDEs are rigorously presented and an existence result is proven.

2.1 Markovian diffusion processes

I now recall some definitions, give new ones and set the notation. Let $\Sigma \subseteq \mathbb{R}^n$ and $\mathcal{B}(\Sigma)$ the associated Borel σ -algebra. Let $(\Omega, \mathcal{F}, P)$ a general probability space. Given $x(s, \omega)$ a Σ -valued random process from $I_0 = [t_0, t_1)$ and $(\Omega, \mathcal{F})$, let us denote by

$$P(C \mid x(s_1), \dots, x(s_m)),$$

the conditional probability of $C \in \mathcal{F}$ given the sigma algebra $\bigvee_{i=1}^m \sigma(x(s_i))$.

Definition 2.1.1. *A stochastic process $x(\cdot, \cdot)$ satisfies the Markov property if there exists a function $p : I_0 \times \Sigma \times I_0 \times \mathcal{B}(\Sigma) \rightarrow \mathbb{R}$ such that:*

1. *for all t, s, B the function $x \mapsto p(t, x, s, B)$ is borel measurable on Σ ,*

2. for all t, x, s the function $A \mapsto p(t, x, s, B)$ is a probability measure on $(\Omega, \mathcal{F})$
3. the Chapman-Kolmogorov equation holds for all $s, t, r \in I_0$, that is if $t < r < s$ then

$$p(t, x, s, B) = \int_{\Sigma} p(r, y, s, B) p(t, x, r, dy). \quad (2.1)$$

Furthermore, for all $r, s \in I_0$ where r, s and for all $B \in \mathcal{B}(\Sigma)$ then

$$P(x(s) \in B \mid \mathcal{F}_r^x) = p(r, x(r), s, B), \quad (2.2)$$

where $\mathcal{F}_r^x = \sigma(x(l) : l \in [t_0, r])$.

The function p is called *Markov Transition Kernel*. We shall see a Markov transition kernel as the probability that the system starting from x at time t will be in B at time s . This heuristic interpretation clarifies the following notation

$$E_{tx}\phi(x(s)) = \int_{\Sigma} \phi(y) p(t, x, s, dy), \quad (2.3)$$

for some real valued borel-measurable function ϕ .

Given a Markov process $x(\cdot, \cdot)$ we can define a family of linear operators associated to it. Let $t < s$, hereafter all time indices will always be in I_0 , and define

$$T_{ts}\phi(x) = \int_{\Sigma} \phi(y) p(t, x, s, dy) = E_{tx}\phi(x(s)). \quad (2.4)$$

Integrability assumptions on ϕ vary from case to case. For now, we can take ϕ to be bounded. Because of Chapman-Kolmogorov equation (2.1) the family $(T_{ts})_{t,s \in I_0}$ satisfies the property

$$T_{tr}[T_{rs}\phi] = T_{ts}\phi \quad (2.5)$$

for all $t < r < s$. This family of linear operators defines another operator, the *backward evolution operator*. Let $A : \{\Phi : I_0 \times \Sigma \rightarrow \mathbb{R}\} \rightarrow \mathbb{R}$ such that

$$A\Phi(t, x) = \lim_{h \rightarrow 0+} \frac{E_{tx}\Phi(t+h, x(t+h)) - \Phi(t, x)}{h}, \quad (2.6)$$

provided that the limit exists. We define $\mathcal{D}(A)$ as the space of functions such that limit (2.6) exists.

Proposition 2.1.1. *Let A as before, then for all $\Phi \in \mathcal{D}(A)$ the following hold:*

1. $\Phi, \frac{\partial \Phi}{\partial t}$ and $A\Phi$ are continuous.
2. For all $t, s \in \bar{I}_0$, $t < s$ then

$$E_{tx}|\Phi(s, x(s))| < +\infty, E_{tx} \int_t^s |A\Phi(r, x(r))| dr < +\infty.$$

3. *Dynkin's formula holds for all $t < s$ that is:*

$$E_{tx}\Phi(s, x(s)) - \Phi(t, x) = E_{tx} \int_t^s A\Phi(r, x(r)) dr, \quad \forall t, s \in I_0, t < s. \quad (2.7)$$

Dynkin's formula can be proved in different instances, subject to the nature of the random process. We will see that it is a natural consequence of Itô's formula for continuous state space processes.

If the random process x is autonomous (time-homogeneous), then the linear operator family is a semigroup. Recall that a Markov process is homogeneous if for all $t < s$ in I_0 then

$$p(t, x, s, B) = p(0, x, s - t, B).$$

If so, by denoting $T_s = T_{0s}$, property (2.5) is given by

$$\begin{aligned} T_{s+r}\phi(x) &= \int_{\Sigma} \phi(y) p(0, x, s + r, dy) \\ &= \int_{\Sigma} \phi(y) \int_{\Sigma} p(r, z, r + s, dy) p(0, x, r, dz) \\ &= \int_{\Sigma} \int_{\Sigma} \phi(y) p(r, z, r + s, dy) p(0, x, r, dz) \\ &= \int_{\Sigma} \int_{\Sigma} \phi(y) p(0, z, s, dy) p(0, x, r, dz) \\ &= \int_{\Sigma} T_s \phi(z) p(0, x, r, dz) \\ &= T_r [T_s \phi(x)]. \end{aligned}$$

While the backward evolution operator analogous is called the *generator* and is defined as

$$G\phi(x) = - \lim_{h \rightarrow 0^+} \frac{T_h \phi(x) - \phi(x)}{h} \quad (2.8)$$

with $D(G)$ being as $\mathcal{D}(A)$ before. It is worth noting that, formally, the following equality holds:

$$A\Phi = \frac{\partial \Phi}{\partial t} - G\Phi(t, \cdot). \quad (2.9)$$

This relation links the two operators and the autonomous to the non-autonomous case. We now turn our attention to a subset of Markov processes: diffusion processes. A diffusion process is a Markov process whose paths are continuous. As Varadhan writes [10] "we would like to identify them through their infinitesimal means and infinitesimal covariance", more formally.

Definition 2.1.2. *A diffusion process $x : \bar{I}_0 \times \Omega \rightarrow \Sigma$ is an almost surely continuous Markov process with Markov transition kernel p such that:*

1. For every $\epsilon > 0$ then

$$\lim_{h \rightarrow 0^+} \int_{|x-y| > \epsilon} p(t, x, t+h, dy) = 0 \quad (2.10)$$

2. There exist functions $a_{ij}(t, x), f_{ij}(t, x)$ for $(t, x) \in \overline{Q}_0$ and $i, j = 1, \dots, n$ such that for every $\epsilon > 0$ then

$$\lim_{h \rightarrow 0^+} \int_{|x-y| \leq \epsilon} (y_i - x_i) p(t, x, t+h, dy) = f_i(t, x), \quad (2.11)$$

and

$$\lim_{h \rightarrow 0^+} \int_{|x-y| \leq \epsilon} (y_i - x_i)(y_j - x_j) p(t, x, t+h, dy) = a_{ij}(t, x). \quad (2.12)$$

These limits are intended uniformly.

Functions $f = (f_1, \dots, f_n)$ and $a = (a_{ij})_{ij}$ are respectively called local drift and local covariance coefficients.

How does the backward evolution operator, and the generator in the autonomous case, adapt to this situation? To answer this question we reduce our problem to a stochastic differential one by relying on the differential structure of a diffusion process. Given the local drift and covariance f, a of a diffusion process x we claim that it satisfies

$$dx(s) = f(s, x(s))ds + \sqrt{a}(s, x(s))dw(s) \quad (2.13)$$

Clearly, we have to impose further conditions of the stochastic differential equation's coefficients to ensure the existence of a solution. In particular, we want those coefficients to be Lipschitz and sub-linearly growing with respect to the second variable. In equation (2.13) we define the square root of a as a function $\sqrt{a} = \sigma$ such that $\sigma(t, x) \cdot \sigma'(t, x) = a(t, x)$. We recall that under existence hypothesis for every $\Phi \in C^{1,2}(\overline{Q}_0)$ Itô's formula holds, that is:

$$d\Phi(s, x(s)) = \Phi_s(s, x(s))ds + \sum_{i=1}^n \Phi_{x^i}(s, x(s))dx^i(s) + \frac{1}{2} \sum_{i,j=1}^n \Phi_{x^i x^j}(s, x(s))d[x, x]^{ij}(s), \quad (2.14)$$

where $[x, y]$ is the covariation of processes x and y . Recall that this relation has always to be intended in integral form, that is:

$$\begin{aligned} \Phi(s, x(s)) &= \Phi(t, x) + \int_t^s \Phi_s(r, x(r)) dr \\ &\quad + \sum_{i=1}^n \int_t^s \Phi_{x^i}(r, x(r)) dx^i(r) + \frac{1}{2} \sum_{i,j=1}^n \int_t^s \Phi_{x^i x^j}(r, x(r)) d[x, x]^{ij}(r). \end{aligned} \quad (2.15)$$

Via this relation, we can reconstruct Dynkin's formula in this setting. By defining the operator A as in (2.7) we have

$$\begin{aligned}
\Phi(s, x(s)) &= \Phi(t, x) + \int_t^s \Phi_s(r, x(r)) dr \\
&+ \sum_{i=1}^n \left[\int_t^s \Phi_{x^i}(r, x(r)) f_i(r, x(r)) dr + \sum_{j=1}^n \int_t^s \Phi_{x^i}(r, x(r)) \sigma_{ij}(r, x(r)) dw^j(r) \right] \\
&+ \frac{1}{2} \sum_{i,j=1}^n \sum_{l=1}^n \int_t^s \Phi_{x^i x^j}(r, x(r)) \sigma_{il}(r, x(r)) \sigma_{jl}(r, x(r)) dr \\
&= \Phi(t, x) + \int_t^s \Phi_s(r, x(r)) + D_x \Phi \cdot f(r, x(r)) + \frac{1}{2} D_x^2 \Phi \cdot a(r, x(r)) dr \\
&+ \int_t^s D_x \Phi \cdot \sigma(r, x(r)) dw(r),
\end{aligned}$$

but we see that the last (stochastic) integral can be seen as a martingale. In particular, if we take Φ to have polynomial growth of some order m , i.e.

$$|\Phi(t, x)| \leq K(1 + |x|^m) \quad \forall (t, x) \in \overline{Q}_0, \quad (2.16)$$

then $D_x \Phi \cdot \sigma \in \mathbb{L}^2(I_0)$, where:

$$\mathbb{L}^2(I_0) = \left\{ x : I \times \Omega \rightarrow \Sigma \mid E \int_I |x(s)|^2 ds < \infty \right\},$$

and therefore its stochastic integral is a martingale (with respect to the canonical filtration associated to the Brownian motion w). Therefore, if we take the (conditional) expectation we get

$$E_{tx} \Phi(s, x(s)) = \Phi(t, x) + E_{tx} \int_t^s \Phi_s(r, x(r)) + D_x \Phi \cdot f(r, x(r)) + \frac{1}{2} D_x^2 \Phi \cdot a(r, x(r)) dr. \quad (2.17)$$

It is now coherent to define the operator $A : C_p^{1,2}(\overline{Q}_0) \rightarrow \mathbb{R}$ as:

$$A\Phi(r, x(r)) = \Phi_s(r, x(r)) + D_x \Phi \cdot f(r, x(r)) + \frac{1}{2} D_x^2 \Phi \cdot a(r, x(r)), \quad (2.18)$$

where $C_p^{1,2}(I)$ is the family of functions g from I into $\mathbb{R}$ such that $g, g_s, g_{x_i}, g_{x_i x_j}$ are continuous with polynomial growth.

Remark. *Be careful that the stochastic integral*

$$\int_t^s D_x \Phi \cdot \sigma(r, x(r)) dw(r)$$

is a martingale because x satisfies

$$E_{tx} |x(r)|^m \leq C_m(1 + |x|^m) \quad \forall r \in I_0,$$

as it is solution of the SDE 2.13.^[1]

^[1]This is a standard result in SDE theory.

Consequently, the generator G of the time-homogeneous case is defined by

$$G\Phi(x) = -\frac{1}{2} \sum_{i,j=1}^n a_{ij}(x) \Phi_{x_i x_j}(x) - \sum_{i=1}^n f_i(x) \Phi_{x_i}(x). \quad (2.19)$$

2.2 Markov control processes

So far we talked about Markov processes without specifying any kind of control. A *control process* is any stochastic process $u : \Omega \rightarrow U$, where U is the control space, that influences the evolution of the random process $x(\cdot, \cdot)$. Formally, let $Q = I_0 \times O$ and u as before and define

$$\begin{cases} dx(r) = f(r, x(r), u(r))dr + \sigma(r, x(r), u(r))dw(r) & r \in I_0 \\ x(t) = x, \end{cases} \quad (2.20)$$

where $U \subset \mathbb{R}^m$ closed, $f, \sigma \in C(\overline{Q}_0 \times U)$, $f(\cdot, \cdot, v), \sigma(\cdot, \cdot, v)$ belong to $C^1(\overline{Q}_0)$ for all $v \in U$, such that there exists a $C > 0$ such that:

$$|f_t| + |f_x| \leq C, \quad |\sigma_t| + |\sigma_x| \leq C, \quad (2.21)$$

$$|f(t, x, v)| \leq C(1 + |x| + |v|), \quad (2.22)$$

$$|\sigma(t, x, v)| \leq C(1 + |x| + |v|). \quad (2.23)$$

We can relax the assumption by imposing Lipschitz condition on t and x for every fixed v . Furthermore, we assume u to be *admissible*, that is

$$E \int_t^{t_1} |u(s)|^m ds < \infty \quad \forall m \in \mathbb{N}. \quad (2.24)$$

Note that it is implied by U being compact. Under these hypotheses, equation (2.20) has a unique ^[2] solution. Where does optimality play its role? We define running and terminal costs L, Ψ , both continuous and satisfying

$$|L(s, x, v)| \leq C(1 + |x|^k + |v|^k), \quad (2.25)$$

$$|\Psi(s, x)| \leq C(1 + |x|^k) \quad (2.26)$$

for suitable $C, k > 0$. We also define τ to be the exit time of $(s, x(s))$ from Q . Finally, we define the final objective function

$$J(t, x; u) = E_{tx} \left\{ \int_t^\tau L(s, x(s), u(s)) ds + \Psi(\tau, x(\tau)) \right\} \quad (2.27)$$

for every initial condition $(t, x) \in Q$ and control u . We aim to minimize this payoff, that is:

^[2]We intend uniqueness as being indistinguishable.

$$\inf_{u \in \mathcal{U}} J(t, x; u).$$

This formulation is not mathematically rigorous enough since we are not mentioning the probability space in which the system takes place; let us further specify optimality. We begin by defining an infimum criterion with respect to a probability space, or more formally a *probability system*, and then we'll take the infimum over all probability systems.

Definition 2.2.1. *A reference probability system is a tuple $\nu = (\Omega, \{\mathcal{F}_s\}, P, \omega)$ such that:*

- a) $(\Omega, \mathcal{F}_{t_1}, P)$ is a probability space,
- b) $\{\mathcal{F}_s\}$ is a filtration on Ω ,
- c) w is an $\mathcal{F}$ -adapted Brownian motion on $[t, t_1]$.

We denote with $\mathcal{A}_{t\nu}$ the collection of all $\mathcal{F}$ -progressively measurable (that is $(\mathcal{B}([t, s]) \times \mathcal{F}_s)$ -adapted), U valued processes u such that condition (2.24) holds on $[t, t_1]$.

We now define

$$V_\nu = \inf_{u \in \mathcal{A}_{t\nu}} J(t, x; u), \quad (2.28)$$

and

$$V_{PM} = \inf_\nu V_\nu. \quad (2.29)$$

Equation (2.28) and (2.29) respectively define ν -*optimality* and *optimality* for those control that satisfy them. We adapt the definition of operator A to this situation incorporating in it every element of the control space v , getting the functional:

$$A^v \Phi = \Phi_t + \sum_{i=1}^n f_i(t, x, v) \Phi_{x_i} + \frac{1}{2} \sum_{i,j=1}^n a_{ij}(t, x, v) \Phi_{x_i x_j}, \quad \Phi \in C_p^{1,2}(\overline{Q}_0). \quad (2.30)$$

Where, again, $a = \sigma \sigma'$. As we did in the deterministic case, we provide a heuristic derivation of the Hamilton-Jacobi-Bellman equation (the verification theorem), to formally prove it later. Let us suppose that $O = \mathbb{R}^n$, then J is

$$J(t, x; u) = \int_t^{t_1} L(s, x(s), u(s)) ds + \Phi(t_1, x(t_1)). \quad (2.31)$$

By the dynamic programming principle, for every $h < t_1 - t$ then

$$V(t, x) = \inf_{u \in \mathcal{A}} E_{tx} \left\{ \int_t^{t+h} L(s, x(s), u(s)) ds + V(t+h, x(t+h)) \right\}.$$

If we take the constant control $u \equiv v$, then by Dynkin's formula we get

$$\begin{aligned}
0 &\leq E_{tx} V(t+h, x(t+h)) - V(t, x) + E_{tx} \int_t^{t+h} L(s, x(s), v) ds \\
&= E_{tx} \int_t^{t+h} A^v V(s, x(s)) ds + E_{tx} \int_t^{t+h} L(s, x(s), v) ds.
\end{aligned}$$

Dividing by h and taking the limit for $h \rightarrow 0^+$:

$$0 \leq A^v V(t, x) + L(t, x, v).$$

If we take u^* to be optimal, then the equality holds:

$$A^{u^*} V(t, x) + L(t, x, u^*(t)) = 0.$$

We now present the verification theorem rigorously. Let us define the Hamiltonian for this problem. For every $(t, x) \in \bar{Q}_0$, $p \in \mathbb{R}^n$ and $A \in \mathcal{S}_+^n$ (set of symmetric, non-negative definite $n \times n$ matrices) we define

$$\mathcal{H}(t, x, p, A) = \sup_{v \in U} \left\{ -f(t, x, v) \cdot p - \frac{1}{2} \text{tr} [a(t, x, v) \cdot A] - L(t, x, v) \right\} \quad (2.32)$$

where for matrices $A, B \in \mathbb{R}^{n \times n}$ we have

$$\text{tr}(AB) = \sum_{i,j=1}^n A_{ij} B_{ji},$$

which is equal to $\sum_{i,j=1}^n A_{ij} B_{ij}$ for symmetric matrices.

We can now state the verification theorem using the Hamiltonian defined in (2.32).

Theorem 2.2.1. *Let $W \in C^{1,2}(Q) \cap C_p(\bar{Q})$ such that:*

$$-\frac{\partial W}{\partial t}(t, x) + \mathcal{H}(t, x, D_x W, D_x^2 W) = 0, \quad \forall (t, x) \in Q, \quad (2.33)$$

$$W(t, x) = \Phi(t, x), \quad \forall (t, x) \in \partial Q. \quad (2.34)$$

Then:

1. *for any system ν , initial condition $(t, x) \in Q$ and any $u \in \mathcal{A}_{t\nu}$ then*

$$W(t, x) \leq J(t, x; u).$$

2. *If there exists $\nu^* = (\Omega^*, \{\mathcal{F}_s^*\}, P^*, w^*)$ and $u^* \in \mathcal{A}_{t\nu^*}$ such that*

$$\begin{aligned}
u^*(s) \in \arg \min_{v \in U} &\left\{ f(s, x^*(s), v) \cdot D_x W(s, x^*(s)) \right. \\
&\left. + \frac{1}{2} \text{tr} [a(s, x^*(s), v) \cdot D_x^2(s, x^*(s))] + L(s, x^*(s), v) \right\}
\end{aligned}$$

for almost all $(s, \omega) \in [t, \tau^*] \times \Omega^*$, then

$$V_{PM}(t, x) = J(t, x; u^*).$$

Proof. We assume O to be bounded and $W \in C^{1,2}(\overline{Q})$. Because of (2.33), for all $s \in [t, \tau]$

$$0 \leq A^{u(s)}W(s, x(s)) + L(s, x(s), u(s)). \quad (2.35)$$

Because of Itô

$$W(\tau, x(\tau)) - W(t, x) = \int_t^\tau A^{u(s)}W(s, x(s)) ds + \int_t^\tau D_x \Phi(s, x(s)) \cdot \sigma(s, x(s), u(s)) dw(s).$$

Because of estimates on SDE solution, the last stochastic integral is a $\mathcal{F}_s$ -martingale. Then if we take the expectation E_{tx} we get:

$$\begin{aligned} 0 &\leq E_{tx} \int_t^\tau A^{u(s)}W(s, x(s)) ds + E_{tx} \int_t^\tau L(s, x(s), u(s)) ds \\ &= E_{tx} [W(\tau, x(\tau)) - W(t, x)] - E_{tx} \int_t^\tau D_x \Phi(s, x(s)) \cdot \sigma(s, x(s), u(s)) dw(s) \\ &\quad + E_{tx} \int_t^\tau L(s, x(s), u(s)) ds \\ &= -W(t, x) + E_{tx} \left\{ \int_t^\tau L(s, x(s), u(s)) ds + W(\tau, x(\tau)) \right\} \\ &= -W(t, x) + J(t, x; u). \end{aligned}$$

If O is unbounded we define for every $\rho > 0$ such that $\rho^{-1} < t_1 - t_0$ the sets

$$O_\rho = O \cap \left\{ |x| < \rho \mid d(x, \partial O) > \frac{1}{\rho} \right\}, \quad Q_\rho = [t_0, t_1 - \rho^{-1}] \times O_\rho,$$

and τ_ρ the exit time from Q_ρ . Q_ρ is bounded, and $W \in C^{1,2}(\overline{Q}_\rho)$, thus

$$W(t, x) \leq E_{tx} \left\{ \int_t^{\tau_\rho} L(s, x(s), u(s)) ds + W(\tau_\rho, x(\tau_\rho)) \right\}. \quad (2.36)$$

We now take $\rho \rightarrow +\infty$ and get the thesis. We have convergence in probability for $\tau_\rho \xrightarrow{\rho \rightarrow +\infty} \tau$. We prove uniform integrability of the rhs, hence L^1 convergence. We have:

$$\begin{aligned} E_{tx} \int_t^{\tau_\rho} |L(s, x(s), u(s))| ds &\leq E_{tx} \int_t^{t_1} |L(s, x(s), u(s))| ds \\ &\leq E_{tx} \int_t^{t_1} \left(1 + |x(s)|^k + |u(s)|^k \right) ds < +\infty \end{aligned}$$

because u is admissible and estimates on SDE solutions. While also have

$$\begin{aligned} E_{tx} |W(\tau_\rho, x(\tau_\rho))|^\alpha &\leq K E_{tx} \left(1 + |x(\tau_\rho)|^k\right)^\alpha \\ &\leq 2^{\alpha-1} K \left(1 + E_{tx} \|x\|^{\alpha k}\right) \leq C, \end{aligned}$$

by estimates on SDE solutions and with $\alpha > \frac{1}{k}$. Therefore, we get

$$\begin{aligned} \lim_{\rho \rightarrow +\infty} E_{tx} \left\{ \int_t^{\tau_\rho} L(s, x(s), u(s)) ds + W(\tau_\rho, x(\tau_\rho)) \right\} \\ = E_{tx} \left\{ \int_t^\tau L(s, x(s), u(s)) ds + W(\tau, x(\tau)) \right\}. \end{aligned}$$

Part b) comes from equality in equation (2.35). $\square$

Remark. *Existence of such optimal control is not tackled in this dissertation, however we refer to the work of Kushner[8] and Fleming and Nisio[5].*

2.3 Stochastic Maximum Principle

The stochastic analogous of Pontryagin's principle is the stochastic maximum principle. It relies on the notion of backward stochastic differential equation, whose solution will provide a necessary condition on the controlled system.

2.3.1 Backward Stochastic Differential Equation

A backward stochastic differential equation (BSDE) is a stochastic differential equation (SDE) where the initial condition is replaced by a final distribution. We start by defining a formal concept of solution and provide a general result about existence and uniqueness.

We work with a filtered probability space $(\Omega, \mathcal{F}, P, \{\mathcal{F}_s\})$ and a Brownian motion (w_s) adapted to the space's filtration. Actually, we assume that the filtration is the natural one associated to w . A BSDE is defined by:

$$\begin{cases} -dy_s = f(s, y_s, z_s)ds - z_s dw_s & \forall s \in [t, t_1] \\ y_{t_1} = \xi, \end{cases} \quad (2.37)$$

where f is real valued and ξ a suitable random variable. Further conditions on f and ξ will be imposed by the existence theorem. As always, the above definition has to be intended in integral form. To do so we have to specify some integrability conditions. Let us recall two functional spaces familiar from SDE theory:

$$\mathbb{S}^2(t, t_1) = \left\{ (X_s)_{t \in [t, t_1]} \mid X_s \in \mathbb{R} \text{ is progressively measurable, } E \left[\sup_{s \in [t, t_1]} |X_s|^2 \right] < +\infty \right\} \quad (2.38)$$

and:

$$\mathbb{H}^2(t, t_1)^n = \left\{ (X_s)_{t \in [t, t_1]} \mid X_s \in \mathbb{R}^n \text{ is progressively measurable, } E \left[\int_t^{t_1} |Y_s|^2 ds \right] < +\infty \right\}. \quad (2.39)$$

We can now define the solution concept of (2.37).

Definition 2.3.1. *A solution of (2.37) is a couple $(y, z) \in \mathbb{S}^2(t, t_1) \times \mathbb{H}^2(t, t_1)^n$ such that:*

$$y_s = \xi + \int_s^{t_1} f(r, y_r, z_r) dr - \int_s^{t_1} z_r dw_r, \quad \forall s \in [t, t_1]. \quad (2.40)$$

Some measurability and integrability conditions are necessary for equation (2.40) to make sense; indeed existence of a solution will be obtained through a classical fixed point method, which will rise from Banach-Cacciopoli's theorem, invocable thanks to a Lipschitz condition on f . We denote by m the Lebesgue measure on $[t, t_1]$. We first need some preliminary results.

Theorem 2.3.1. *Let $f : \Omega \times [t, t_1] \times \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ such that $f(\cdot, \cdot, y, z)$ is progressively measurable for all $(y, z) \in \mathbb{R}^{n+1}$, $f(\cdot, 0, 0) \in \mathbb{H}^2(t, t_1)^1$ and there exists $C > 0$ such that*

$$|f(s, y_1, z_1) - f(s, y_2, z_2)| \leq C(|y_1 - y_2| + |z_1 - z_2|), \quad \forall y_1, y_2, z_1, z_2, \quad m \otimes P \text{ a.s.} \quad (2.41)$$

Then, for every $\xi \in L^2$ the BSDE (2.37) has a unique solution.

Proof. We use the completeness of the space $(\mathbb{S}(t, t_1) \times \mathbb{H}^2(t, t_1)^n, \|\cdot\|_\beta)$, where

$$\|(y, z)\|_\beta = \left(E \left[\int_t^{t_1} e^{\beta s} (|y_s|^2 + |z_s|^2) ds \right] \right)^{1/2}.$$

We denote by $(X, \|\cdot\|)$ the previous Banach space. We construct the map $\Phi : X \rightarrow X : (u, v) \mapsto (y, z)$, where processes y and z are defined as follows. We define

$$M_s = E \left[\xi + \int_0^{t_1} f(r, u_r, v_r) dr \mid \mathcal{F}_s \right]. \quad (2.42)$$

It is a square integrable martingale because

$$\begin{aligned} E \left\{ E^2 \left[\xi + \int_t^{t_1} f(r, u_r, v_r) dr \mid \mathcal{F}_s \right] \right\} &\leq C_0 E^2 \xi + C_1 E^2 \int_t^{t_1} f(r, 0, 0) dr \\ &\quad + C_2 E^2 \int_t^{t_1} f(r, u_r, v_r) - f(r, 0, 0) dr \\ &\quad + C_3 E \xi^2 E \int_t^{t_1} f(r, 0, 0)^2 dr \\ &\quad + C_4 E \xi^2 E \int_t^{t_1} (f(r, u_r, v_r) - f(r, 0, 0))^2 dr, \end{aligned}$$

which is finite because of the assumptions and given $(u, v) \in X$. Therefore, by Itô's martingale representation theorem there exists a unique $z \in \mathbb{H}^2(t, t_1)^{n[3]}$ and $m_t \in L^2$ such that

$$M_s = m_t + \int_t^s z_r dw_r. \quad (2.43)$$

We define (y, z) as z being the unique process of the martingale representation of M_s while y to be

$$y_s = E \left[\xi + \int_s^{t_1} f(r, u_r, v_r) dr \mid \mathcal{F}_s \right] = M_s - \int_t^s f(r, u_r, v_r) dr. \quad (2.44)$$

We know that $z \in \mathbb{H}^2$. We show $y \in \mathbb{S}^2$:

$$E \left[\sup_{s \in [t, t_1]} |y_s|^2 \right] \leq C_0 E \left[\sup_{s \in [t, t_1]} |m_s|^2 \right] + C_1 E \left[\sup_{s \in [t, t_1]} \left| \int_t^s f(r, u_r, v_r) dr \right|^2 \right]. \quad (2.45)$$

The second addend is finite by hypothesis, while for the first one holds:

$$\begin{aligned} E \left[\sup_{s \in [t, t_1]} |m_s|^2 \right] &\leq C_0 E m_t^2 + C_1 E^{1/2} m_t^2 E^{1/2} \sup_{s \in [t, t_1]} \left[\int_t^s z_r dw_r \right]^2 \\ &\quad + C_3 E^{1/2} \sup_{s \in [t, t_1]} \left[\int_t^s z_r dw_r \right]^2, \end{aligned}$$

which is finite because Doob's inequality implies

$$E \left[\sup_{s \in [t, t_1]} \int_t^s z_r dw_r \right]^2 \leq 4E \left[\int_t^{t_1} z_r^2 dw_r \right] < +\infty.$$

If we prove Φ to be a contraction we'll have a unique infixed point $(X, \|\cdot\|)$, therefore the thesis.

Let $(U_1, V_1), (U_2, V_2) \in X$, $(X_1, Y_1), (X_2, Y_2)$ their images and $\bar{U}, \bar{V}, \bar{X}, \bar{Y}, \bar{f}_t$ the differences between the respective subscript 1 and 2. By Ito's formula we have

$$\begin{aligned} |\bar{y}_t^2| &= - \int_t^{t_1} \frac{d}{dr} (e^{\beta r} \bar{y}_r^2) dr = - \int_t^{t_1} \beta e^{\beta r} \bar{y}_r^2 + e^{\beta r} \frac{d}{dr} \bar{y}_r^2 dr \\ &= - \int_t^{t_1} \beta e^{\beta r} \bar{y}_r^2 dr - \int_t^{t_1} e^{\beta r} (2\bar{y}_r) d\bar{y}_r - \int_t^{t_1} e^{\beta r} d[\bar{y}, \bar{y}]_r \\ &= - \int_t^{t_1} \beta e^{\beta r} \bar{y}_r^2 dr + 2 \int_t^{t_1} e^{\beta r} \bar{y}_r \bar{f}_r dr - 2 \int_t^{t_1} e^{\beta r} \bar{y}_r dw_r - \int_t^{t_1} e^{\beta r} \bar{z}_r^2 dr \\ &= - \int_t^{t_1} e^{\beta r} (\beta \bar{y}_r^2 - 2\bar{y}_r \bar{f}_r) dr - \int_t^{t_1} e^{\beta r} \bar{z}_r^2 dr - 2 \int_t^{t_1} e^{\beta r} \bar{y}_r \bar{z}_r dw_r. \end{aligned} \quad (2.46)$$

[3]Unique with respect to $\|\cdot\|_{\mathbb{H}}$.

We show that $e^{\beta r} \bar{y}_r \bar{z}_r \in \mathbb{H}^2(t, t_1)$, which will guarantee the integral to vanish under expectation. We have

$$E \left[\left(\int_t^{t_1} e^{2\beta r} \bar{y}_r^2 \bar{z}_r^2 dr \right)^{1/2} \right] \leq \frac{e^{\beta t_1}}{2} E \left[[t_1 - t] \sup_{s \in [t, t_1]} \bar{y}_s^2 + \int_t^{t_1} \bar{z}_r^2 dr \right] < +\infty.$$

Thus, by taking the expectation on (2.46) we have

$$\begin{aligned} E \bar{y}_t^2 + E \left[\int_t^{t_1} e^{\beta r} (\beta \bar{y}_r^2 + \bar{z}_r^2) dr \right] &= 2E \left[\int_t^{t_1} e^{\beta r} \bar{y}_r \bar{f}_r dr \right] \\ &\leq 2CE \left[\int_t^{t_1} e^{\beta r} \bar{y}_r (\bar{u}_r + \bar{v}_r) dr \right] \\ &\leq 2CE \left[\int_t^{t_1} e^{\beta r} \bar{y}_r^2 dr \right] + CE \left[\int_t^{t_1} e^{\beta r} (\bar{u}_r^2 + \bar{v}_r^2) dr \right], \end{aligned}$$

which implies

$$E \left[\int_t^{t_1} e^{\beta r} (\bar{y}_r^2 + \bar{z}_r^2) dr \right] \leq E \left[\int_t^{t_1} e^{\beta r} (\bar{u}_r^2 + \bar{v}_r^2) dr \right].$$

□

A notable case is the one with a generator f linear in y and z . That is, there exists a, b bounded progressively measurable processes valued in $\mathbb{R}$ and $\mathbb{R}^n$ and $c \in \mathbb{H}(t, t_1)^1$ which define:

$$-dy_s = (a_s y_s + z_s b_s + c_s) ds - z_s dw_s, \quad y_{t_1} = \xi. \quad (2.47)$$

Proposition 2.3.2. *The unique solution (y, z) of (2.47) is*

$$\Gamma_s y_s = E \left[\Gamma_{t_1} \xi + \int_s^{t_1} \Gamma_r c_r dr \mid \mathcal{F}_s \right], \quad (2.48)$$

and z_s is defined via the Martingale representation of (2.48). The process Γ is defined by

$$d\Gamma_s = \Gamma_s (a_s ds + b_s dw_s), \quad \Gamma_t = 1.$$

2.3.2 Stochastic Maximum Principle

The stochastic counterpart of Pontryagin's principle is the stochastic maximum principle. We consider the controlled diffusion process x on $\mathbb{R}^n$ defined by

$$dx_s = f(r, x(r), u(r))dr + \sigma(r, x(r), u(r))dw(r), \quad (2.49)$$

and let the functional to maximized be

$$J(t, x; u) = E_{tx} \left\{ \int_t^{t_1} L(s, x(s), u(s)) ds + \Phi(t_1, x(t_1)) \right\}, \quad (2.50)$$

where the running and terminal cost satisfy the usual assumptions. We aim at minimizing the functional J over all controls in a given admissible system, that is:

$$\inf_{u \in \mathcal{A}_{t\nu}} J(t, x; u).$$

We consider again the value function related to this problem, that is a function V , with suitable smoothness conditions, such that:

$$-\frac{\partial V}{\partial t} + \mathcal{H}(t, x, D_x V, D_x^2 V) = 0, \quad (2.51)$$

where

$$\mathcal{H}(s, x, v, p, A) = -f(s, x, v) \cdot p - \frac{1}{2} \text{tr} [\sigma \sigma'(s, x, v) \cdot A] - L(s, x, v). \quad (2.52)$$

As in the determinist case, under the assumption of optimality the value function V will define the solution of a differential equation, more precisely a BSDE. The backward stochastic differential equation will be an analogous of the deterministic adjoint equation

$$\begin{cases} \dot{p}(s) = D_x H(s, x(s), u(s), p(s)) \\ p(t_1) = D\psi(x(t_1)), \end{cases}$$

as stated in (1.3.1). In particular, we will need the functional

$$\mathcal{G}(t, x, v, y, z) = -f(s, x, v) \cdot y - \text{tr} [\sigma'(s, x, v) \cdot z] - L(s, x, v). \quad (2.53)$$

This functional will define the BSDE in the next theorem.

Theorem 2.3.3. *Let u^* be an optimal control and x^* the corresponding diffusion process, and the value function $V \in C^{1,3}(O) \cap C(\overline{O})$. Then V satisfies*

$$\mathcal{H}(s, x_s^*, u_s^*, D_x V(s, x_s^*), D_x^2 V(s, x_s^*)) = \sup_{v \in U} \mathcal{H}(s, x_s^*, v, D_x V(s, x_s^*), D_x^2 V(s, x_s^*)), \quad (2.54)$$

and the pair $(y_s, z_s) = (D_x V(s, x_s^*), D_x^2 v(s, x_s^*) \sigma(s, x_s^*, u_s^*))$ solves the BSDE

$$-dy_s = D_x \mathcal{G}(s, x_s^*, u_s^*, y_s, z_s) ds - z_s dw_s, \quad (2.55)$$

with final condition

$$y_{t_1} = D_x \Psi(t_1, x_{t_1}). \quad (2.56)$$

Proof. We drop the *. We consider the optimal system over which the control is defined. Since u is optimal we have

$$\begin{aligned} V(s, x(s)) &= E_{sx(s)} \left[\int_s^{t_1} L(r, x(r), u(r)) dr + \Psi(t_1, x(t_1)) \right] \\ &= E_{tx} \left[\int_t^{t_1} L(r, x(r), u(r)) dr + \Psi(t_1, x(t_1)) \right] - \int_t^s L(r, x(r), u(r)) dr. \end{aligned}$$

We then apply Itô's formula to $V(s, x(s))$ and get:

$$\begin{aligned} V(s, x(s)) &= \int_t^{t_1} \partial_s V(r, x(r)) + D_x V(r, x(r)) f(r, x(r), u(r)) \\ &\quad + \frac{1}{2} tr [\sigma \sigma'(r, x(r), u(r)) D_x^2 V(r, x(r))] dr \\ &\quad + \int_t^{t_1} D_x V(r, x(r)) \sigma(r, x(r), u(r)) dw_r + V(t, x) \\ &= \int_t^{t_1} -f(r, x(r), u(r)) dr + \int_t^{t_1} \alpha_r dw_r + V(t, x), \end{aligned} \tag{2.57}$$

then taking the expectation and identifying the terms we get

$$-\frac{\partial V}{\partial t} + \mathcal{H}(s, x(s), u(s), D_x V(s, x(s)), D_x^2 V(s, x(s))) = 0. \tag{2.58}$$

We then prove the BSDE using the continuity of triple spacial derivatives of the value functions and the fact that (2.58) at s has maximum in $x(s)$, which implies

$$\frac{\partial}{\partial x} \left(\frac{\partial V}{\partial t}(s, x) - \mathcal{H}(s, x, u(s), D_x V(s, x), D_x^2 V(s, x)) \right) |_{x=x(s)} = 0.$$

We compute the derivative and get:

$$\begin{aligned} &\frac{\partial^2 V}{\partial x \partial t}(s, x(s)) + D_x f(s, x, u(s))|_{x=x(s)} D_x V(s, x(s)) + f(s, x(s), u(s)) D_x^2 V(s, x(s)) \\ &\quad + \frac{1}{2} tr [D_x(\sigma \sigma')(s, x, u(s))|_{x=x(s)} \cdot D_x^2 V(s, x(s))] \\ &\quad + \frac{1}{2} tr [\sigma \sigma'(s, x(s), u(s)) \cdot D_x^3 V(s, x(s))] + D_x L(s, x(s), u(s)) \\ &= \frac{\partial^2 V}{\partial x \partial t}(s, x(s)) + f(s, x(s), u(s)) D_x^2 V(s, x(s)) \\ &\quad + \frac{1}{2} tr [\sigma \sigma'(s, x(s), u(s)) \cdot D_x^3 V(s, x(s))] \\ &\quad + f(s, x(s), u(s)) D_x V(s, x(s)) \\ &\quad + tr [D_x \sigma'(s, x(s), u(s)) \cdot D_x^2 V(s, x(s)) \sigma(s, x(s), u(s))] \\ &\quad + D_x L(s, x(s), u(s)), \end{aligned}$$

which implies

$$\begin{aligned}
& \frac{\partial^2 V}{\partial x \partial t}(s, x(s)) + f(s, x(s), u(s)) D_x^2 V(s, x(s)) \\
& + \frac{1}{2} \text{tr} [\sigma \sigma'(s, x(s), u(s)) \cdot D_x^3 V(s, x(s))] \\
& = -D_x [\mathcal{G}(s, x(s), u(s), D_x V(s, x(s)), D_x^2 V(s, x(s)) \sigma(s, x(s), u(s)))] .
\end{aligned} \tag{2.59}$$

Equation (2.59) implies that y_s is the unique solution to the BSDE (2.55):

$$\begin{aligned}
-dy_s &= -d[D_x v(s, x(s))] \\
&= -\frac{\partial^2 V}{\partial x \partial t}(s, x(s))ds - f(s, x(s), u(s)) D_x^2 V(s, x(s))ds \\
&\quad - D_x^2 V(s, x(s)) \sigma(s, x(s), u(s)) dw_s \\
&\quad + \frac{1}{2} \text{tr} [\sigma \sigma'(s, x(s), u(s)) \cdot D_x^3 V(s, x(s))] ds \\
&= D_x [\mathcal{G}(s, x(s), u(s), D_x V(s, x(s)), D_x^2 V(s, x(s)) \sigma(s, x(s), u(s)))] ds \\
&\quad - D_x^2 V(s, x(s)) \sigma(s, x(s), u(s)) dw_s,
\end{aligned}$$

while

$$V(t_1, x(t_1)) = E_{t_1 x(t_1)} \left[\int_{t_1}^{t_1} L(r, x(r), u(r)) dr + \Psi(t_1, x(t_1)) \right] = \Psi(t_1, x(t_1)),$$

which gives

$$D_x V(t_1, x(t_1)) = D_x \Psi(t_1, x(t_1)).$$

□

Chapter 3

Viscosity Solutions

This chapter introduces the theory and application of viscosity solutions within the context of optimal control problems. It addresses the challenge of non-differentiable value functions, which often arise in practical scenarios, making classical solution methods inapplicable. Viscosity solutions provide a framework to handle these cases by weakening the differentiability requirements while still ensuring a well-defined and unique solution to the Hamilton-Jacobi-Bellman (HJB) equations. The chapter begins with an introduction to the concept of viscosity solutions, motivated by an explicit computation of a non-differentiable value function. The method of vanishing viscosity is introduced as a historical motivation for this new notion of solution; perturbing the system to convert a fully non-linear first-order PDE into a semi-linear one then taking the limit of these solutions as the perturbation parameter approaches zero, a viscosity solution is obtained. The formal definition of viscosity solutions is provided, establishing conditions under which a function is considered a viscosity subsolution or supersolution. This section emphasizes the importance of these definitions for ensuring that the solution adheres to the dynamic programming principle even when the value function is not smooth. The chapter also explores the standard approach to viscosity solutions for partial differential operators, highlighting the maximum principle and ellipticity conditions that the operator must satisfy. Detailed discussions on the continuity properties of the value function are included, proving that under specific assumptions, the value function remains bounded and uniformly continuous. Further, the extension of Pontryagin's principle to non-differentiable settings using viscosity solutions is discussed. The classical adjoint variable is extended to accommodate non-smooth points by defining superdifferentials and subdifferentials. This ensures that the viscosity solution framework remains robust and consistent for solving optimal control problems where traditional differentiable approaches fail.

3.0.1 Non-differentiable value functions

Let us consider the calculus of variation problem:

$$\inf_{x \in Lip([0,1];[-1,1])} \int_t^{t_1} 1 + \frac{1}{4}(\dot{x}(s))^2 ds, \quad (3.1)$$

where $Lip(I; U)$ is the collection of Lipschitz continuous functions from I to U . We can see this as an optimal control problem where $I_0 = [0, 1]$, $\Psi \equiv 0$, and $L(t, x, v) = 1 + \frac{v^2}{4}$. The Hamiltonian related to this problem is

$$H(t, x, p) = \max_{v \in \mathbb{R}} \left\{ -v \cdot p - 1 - \frac{1}{4}v^2 \right\}.$$

We can explicitly compute the Hamiltonian and get

$$H(t, x, p) = p^2 - 1.$$

Then the Hamilton-Jacobi equations read

$$\begin{cases} \dot{x}^*(s) = -H_p(s, x^*(s), p^*(s)) = 2p^*(s) \\ \dot{p}^*(s) = H_x(s, x^*(s), p^*(s)) = 0, \end{cases}$$

therefore, we get:

$$\dot{x}(s)^* = 2p^*, \quad s \in [0, 1],$$

thus a linear $x^*(\cdot)$ and a constant optimal control u^* . We have

$$J(t, x; v) = \left(1 + \frac{v^2}{4}\right) (\tau - t),$$

where

$$\tau = \begin{cases} 1 & \text{or} \\ t + \frac{1-x}{v} & \text{or} \\ t - \frac{1+x}{v}, \end{cases}$$

therefore, if we maximize J for each values of τ we get $v = 0, 2, -2$ respectively. In order to assign each v a domain we observe that if $|x| < t$ then neither $v = 2$ nor $v = -2$ are enough for $x(\cdot)$ to optimally exit from $[-1, 1]$ before $s = 1$, and thus it is cost minimizing not to move ($v = 0$), while if $x \geq t$ or $x \leq -t$ it is worth and we respectively get $v = 2$ and $v = -2$. Thus,

$$u^* = \begin{cases} 2, & x \geq t, \\ 0, & |x| < t, \\ -2, & x \leq -t, \end{cases}$$

and

$$\tau = \begin{cases} \frac{1-x}{2} + t, & x \geq t, \\ 1, & |x| < t, \\ t + \frac{1+x}{2}, & x \leq -t. \end{cases}$$

Thus,

$$V(t, x) = J(t, x; u^*) = \left(1 + \frac{(u^*)^2}{4}\right) (\tau - t) = \begin{cases} 1 - |x| & |x| \leq t \\ 1 - t & |x| \geq t, \end{cases} \quad (3.2)$$

which is continuous on the whole space, but clearly not differentiable in $|x| = t$.

3.0.2 Vanishing viscosity

We now euristically expose a technique called vanishing viscosity, which is widely used in calculus of variations problems. It will show us one of the origins of the viscosity solution notion. Let us consider the initial value problem

$$\begin{cases} u_t + H(u, Du) = 0, & \mathbb{R}^n \times (0, +\infty) \\ u = g, & \mathbb{R}^n \times \{t = 0\}. \end{cases} \quad (3.3)$$

The method of characteristics shows that there cannot be a smooth solution of the above problem over the whole positive real line, thus a weaker notion of solution is needed. One approach is to use Hopf-Lax solution concept. We are not interested in it, instead we search for solution from similar problems which we already know. Let us perturbate the system as

$$\begin{cases} u_t^\epsilon + H(u^\epsilon, Du^\epsilon) - \epsilon \Delta u^\epsilon = 0, & \mathbb{R}^n \times (0, +\infty) \\ u^\epsilon = g, & \mathbb{R}^n \times \{t = 0\}, \end{cases} \quad (3.4)$$

so that the fully non-linear system in 3.3 becomes a semilinear one, which has a smooth solution, e.g. Evans[4] illustrates the Galerkin approximation method. Since the system (3.3) is some sort of limit version of (3.4) we are tempted to take the limit of its solutions as $\epsilon \rightarrow 0$ and see if, provided convergence, we get the solution. Thus, we take $\epsilon \rightarrow 0$. We expect the solution u^ϵ to lose the bounds on the derivatives, as they strongly depend on the regularization effect of $\epsilon \Delta$. Turns out that many times Ascoli-Arzelà theorem's hypotheses are satisfied, that is $(u^\epsilon)_\epsilon$ is uniformly bounded and equicontinuous, then we have local uniform convergence along a subsequence u^{ϵ_j} . We now use the limit $u \xleftarrow{j \rightarrow +\infty} u^{\epsilon_j}$ as a solution. We know the limit to be continuous but we lack information about its derivatives. We will then verify these information using test functions. Unlike the classical variational weak solution concept, where integration by part plays the central role, we will use the maximum principle to translate the derivatives of u onto the test functions. Let us take $v \in C^\infty(\mathbb{R}^n \times (0, +\infty))$ and suppose that $u - v$ has a strict local maximum at (x_0, t_0) and $u(x_0, t_0) = v(x_0, t_0)$, then

$$(u - v)(x_0, t_0) > (u - v)(x, t),$$

for all (t, x) sufficiently close to (x_0, t_0) . It can be shown that it implies that there exists $J > 0$ such that for all $j > J$ there exists $(x_{\epsilon_j}, t_{\epsilon_j})$ such that

$$(u^\epsilon - v)(x_{\epsilon_j}, t_{\epsilon_j}) \geq (u^\epsilon - v)(x, t),$$

for (x, t) sufficiently close to $(x_{\epsilon_j}, t_{\epsilon_j})$ and such that

$$(x_{\epsilon_j}, t_{\epsilon_j}) \xrightarrow{j \rightarrow +\infty} (x_0, t_0).$$

Because $(u^\epsilon - v)$ has a local maximum at $(x_{\epsilon_j}, t_{\epsilon_j})$ then

$$Du^{\epsilon_j}(x_{\epsilon_j}, t_{\epsilon_j}) = Dv(x_{\epsilon_j}, t_{\epsilon_j}), \quad u_t^{\epsilon_j}(x_{\epsilon_j}, t_{\epsilon_j}) = v(x_{\epsilon_j}, t_{\epsilon_j}),$$

and

$$-\Delta u^{\epsilon_j}(x_{\epsilon_j}, t_{\epsilon_j}) \geq -\Delta v(x_{\epsilon_j}, t_{\epsilon_j}).$$

Therefore,

$$v_t(x_0, t_0) + H(Dv(x_0, t_0), x_0) \xleftarrow{j \rightarrow +\infty} v_t(x_{\epsilon_j}, t_{\epsilon_j}) + H(Dv(x_{\epsilon_j}, t_{\epsilon_j}), x_{\epsilon_j}) \leq \Delta_{\epsilon_j} v(x_{\epsilon_j}, t_{\epsilon_j}) \xrightarrow{j \rightarrow +\infty} 0.$$

Analogous computations can be done for local minimum of $u - v$, obtaining the opposite inequality above.

We can now grasp the intuition behind the following definition.

Definition 3.0.1. *A viscosity solution of 3.3 is a function u bounded and uniformly continuous on $\mathbb{R}^n \times [0, T]$ for all $T > 0$ such that $\forall v \in C^\infty(\mathbb{R}^n \times (0, +\infty))$*

$$v_t(x, t) + H(Dv(x, t), x) \leq 0$$

for all $(x, t) \in \arg \max\{u - v\}$ and

$$v_t(x, t) + H(Dv(x, t), x) \geq 0$$

for all $(x, t) \in \arg \min\{u - v\}$. Furthermore, $u \equiv g$ for $t = 0$.

3.1 Abstract dynamic programming

We now present an abstraction of the dynamic programming principle, which will allow us to define the viscosity solutions of the dynamic programming equation. Let Σ be a closed subset of a Banach space and $\mathcal{C}$ a collection of functions on Σ , closed under addition, that is

$$\phi, \psi \in \mathcal{C} \Rightarrow \phi + \psi \in \mathcal{C}.$$

We consider the family of operators $\{\mathcal{T}_{tr}\}_{t_0 \leq t \leq r \leq t_1}$ such that:

$$\mathcal{T}_{tt}\phi = \phi, \quad \forall \phi \in \mathcal{C}, \tag{3.5}$$

$$\mathcal{T}_{tr}\phi \leq \mathcal{T}_{ts}\psi \text{ if } \phi \leq \mathcal{T}_{rs}\psi, \tag{3.6}$$

and

$$\mathcal{T}_{tr}\phi \geq \mathcal{T}_{ts}\psi \text{ if } \phi \geq \mathcal{T}_{rs}\psi. \tag{3.7}$$

Conditions 3.6 and 3.7 are a weaker version of monotonicity; they imply it together with 3.5. Moreover, they also imply the semigroup property, provided that $\mathcal{T}_{rt} : \mathcal{C} \rightarrow \mathcal{C}$. If the latter holds true, the two conditions are equivalent to monotonicity. The semigroup property

$$\mathcal{T}_{tr}(\mathcal{T}_{rs}\psi) = \mathcal{T}_{ts}\psi, \quad \mathcal{T}_{rs}\psi \in \mathcal{C}, \quad (3.8)$$

will mimic the dynamic programming principle. Indeed, let us consider the classical optimal control problem defined on a bounded set $O \subset \mathbb{R}^n$, which we choose to be $\Sigma = \overline{O}$ and $\mathcal{C} = \mathcal{M}(\Sigma)$, the collection of measurable functions bounded by below. Then, as in chapter 1, we aim at minimizing a functional, we set this functional to be the operator $\mathcal{T}$. Let us define

$$\mathcal{T}_{t,r;u}\psi(x) = \int_t^{\tau \wedge r} L(s, x(s), u(s)) ds + g(\tau, x(\tau))\chi_{\tau < r} + \psi(x(r))\chi_{\tau \geq r},$$

which gives

$$\mathcal{T}_{tr}\psi = \inf_{u \in \mathcal{U}(t,x)} \mathcal{T}_{t,r;u}\psi. \quad (3.9)$$

Under the usual assumption on the running and terminal costs as well as on the control space U , we note that the value function defined in 3.1 is measurable and bounded by below. Therefore, $\mathcal{T}_{rt} : \mathcal{C} \rightarrow \mathcal{C}$ and we can formulate the semigroup property just by asking $\psi \in \mathcal{C}$. It is clear by its definition that the dynamic programming principle is

$$\mathcal{T}_{tt_1}\psi(x) = \mathcal{T}_{tr}(\mathcal{T}_{rt_1}\psi)(x),$$

for $(t, x) \in \overline{Q}$ and $\psi \in \mathcal{C}$. Let us define the value function as

$$V(t, x) = (\mathcal{T}_{tt_1}\psi)(x). \quad (3.10)$$

We now derive the abstract dynamic programming equation. The same procedure as in chapter 1 yields

$$-\frac{1}{h} [\mathcal{T}_{tt+h}V(t+h, \cdot)(x) - V(t, x)] = 0.$$

What happens if we let $h \rightarrow 0$? We ask for the existence of a family of non-linear operators which will play the role of the Hamiltonian. Let $\Sigma' \subset \Sigma$ and $\mathcal{D} \subset C([t_0, t_1] \times \Sigma')$ and $\{\mathcal{G}_t\}_{t \in [t_0, t_1]}$ functions on Σ such that:

$$\lim_{h \rightarrow 0} \frac{1}{h} [\mathcal{T}_{tt+h}V(t+h, \cdot)(x) - V(t, x)] = \frac{\partial}{\partial t}w(t, x) - (\mathcal{G}_tw(t, \cdot))(x) \quad (3.11)$$

for all $w \in \mathcal{D}$, $(t, x) \in Q = [t_0, t_1] \times \Sigma'$. The space $\mathcal{D}$ is such that:

$$\forall w \in \mathcal{D} \text{ the functions } \frac{\partial w}{\partial t}, \mathcal{G}_tw(t, \cdot) \text{ are continuous on } Q \text{ and } w(t, \cdot) \in \mathcal{C} \forall t \in [t_0, t_1], \quad (3.12)$$

and $\mathcal{D}$ is a vector space

$$w, \tilde{w} \in \mathcal{D} \Rightarrow w + \tilde{w} \in \mathcal{D}, \lambda w \in \mathcal{D}. \quad (3.13)$$

The elements of $\mathcal{D}$ are called test functions and $\mathcal{G}_t$ the infinitesimal generator of $\mathcal{T}_{tr}$. Explicit choices of $\mathcal{C}$ and $\mathcal{D}$ will vary from case to case, usually they are chosen to satisfy certain integrability conditions on the functions.

If we require the existence of a test function space and an infinitesimal generator of the semigroup given by the value function, the dynamic programming equation becomes

$$-\frac{\partial}{\partial t}V(t, x) + (\mathcal{G}_t V(t, \cdot))(x) = 0, (t, x) \in Q. \quad (3.14)$$

Then, if a function $V \in \mathcal{D}$ satisfies 3.14 point wise it is called a classical solution of it. Thanks to this reformulation we will be able to weaken this notion of solution and get the viscosity solution of the dynamic programming equation.

We point out that in deterministic optimal control, the infinitesimal generator has the form

$$(\mathcal{G}_t \phi)(x) = H(t, x, D\phi(x)) = \sup_{v \in U} \{-f(t, x, v) \cdot D\phi(x) - L(t, x, v)\},$$

with test functions space $\mathcal{D} = C^1(Q) \cap \mathcal{M}(\overline{Q})$ and $\Sigma' = O$.

3.1.1 Viscosity Solutions

In view of what we saw in previous sections, we give the definition of viscosity solution.

Definition 3.1.1. *Let $W \in C([t_0, t_1] \times \Sigma)$. Then:*

1. *W is a viscosity subsolution of 3.14 in Q if for every $w \in \mathcal{D}$*

$$-\frac{\partial}{\partial t}w(\bar{t}, \bar{x}) + (\mathcal{G}_{\bar{t}}w(\bar{t}, \cdot))(\bar{x}) \leq 0, \quad (3.15)$$

at every

$$(\bar{t}, \bar{x}) \in \arg \max_{(t, x) \in Q} \{(W - w)(t, x)\},$$

and $W(\bar{t}, \bar{x}) = w(\bar{t}, \bar{x})$.

2. *W is a viscosity supersolution of 3.14 in Q if for every $w \in \mathcal{D}$*

$$-\frac{\partial}{\partial t}w(\bar{t}, \bar{x}) + (\mathcal{G}_{\bar{t}}w(\bar{t}, \cdot))(\bar{x}) \geq 0, \quad (3.16)$$

at every

$$(\bar{t}, \bar{x}) \in \arg \min_{(t, x) \in Q} \{(W - w)(t, x)\},$$

and $W(\bar{t}, \bar{x}) = w(\bar{t}, \bar{x})$.

3. W is a viscosity solution if it is both a subsolution and a supersolution.

As a first step after the definition of viscosity solutions, we prove its consistency with the classical notion. To do so we require the operator to satisfy a maximum principle, that is if $\mathcal{G}_t$ is a general operator and

$$\mathcal{D} = \{W \in C([t_0, t_1] \times \Sigma) \mid W_t(t, x), (\mathcal{G}_t W(t, \cdot))(x) \in C(Q)\},$$

then $\mathcal{G}_t$ satisfies the *maximum principle* if

$$\mathcal{G}_t \phi(\bar{x}) \geq \mathcal{G}_t \psi(\bar{x})$$

for every $\bar{x} \in \arg \max\{(\phi - \psi)(x) \mid x \in \Sigma\} \cap \Sigma'$ with $\phi(\bar{x}) = \psi(\bar{x})$. If W is a classical solution of 3.14, then for a $w \in \mathcal{D}$ and $(\bar{t}, \bar{x}) \in \arg \max_{(t,x) \in Q} \{(W - w)(t, x)\}$ and $W(\bar{t}, \bar{x}) = w(\bar{t}, \bar{x})$ we have

$$-\frac{\partial}{\partial t} w(\bar{t}, \bar{x}) + (\mathcal{G}_{\bar{t}} w(\bar{t}, \cdot))(\bar{x}) \leq -\frac{\partial}{\partial t} W(\bar{t}, \bar{x}) + (\mathcal{G}_{\bar{t}} W(\bar{t}, \cdot))(\bar{x}) = 0,$$

Since we asked continuity of time derivatives. For the supersolution recall that:

$$\max\{(\phi - \psi)(x) \mid x \in \Sigma\} = \min\{-(\phi - \psi)(x) \mid x \in \Sigma\}.$$

If $\mathcal{G}_t$ is the infinitesimal generator of a two-parameter semigroup the connection between classical and viscosity solutions is even stronger.

Proposition 3.1.1. *Let $W \in \mathcal{D}$. Then W is a classical solution of 3.14 if and only if it is a viscosity solution of 3.14 in Q .*

Proof. If W is a viscosity solution, since it is also a test function then 3.15 and 3.16 hold for every point $(t, x) \in Q$, which implies that

$$-\frac{\partial}{\partial t} w(t, x) + (\mathcal{G}_t w(t, \cdot))(x) = 0, \quad \forall (t, x) \in Q.$$

If W is a classical solution and prove the subsolution property. Let $w \in \mathcal{D}$ and (t, x) as usual. Since $w \geq W$

$$\begin{aligned} -\frac{\partial}{\partial t} w(t, x) + (\mathcal{G}_t w(t, \cdot))(x) &= -\lim_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{tt+h} w(t+h, \cdot))(x) - w(t, x)] \\ &\leq -\lim_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{tt+h} W(t+h, \cdot))(x) - W(t, x)] \\ &= -\frac{\partial}{\partial t} W(t, x) + (\mathcal{G}_t W(t, \cdot))(x) = 0. \end{aligned}$$

The supersolution property is proven similarly. □

We now prove that the family of linear operators $\mathcal{T}_{tt_1}$ defined in 3.9 defines viscosity solution of the dynamic programming equation 3.14.

Theorem 3.1.2. *Let $\{\mathcal{T}_{tr}\}_{t_0 \leq t \leq r \leq t_1}$ such that 3.5, 3.7, 3.6 and also there exists a vector space $\mathcal{D}$ and another family of operator $\{\mathcal{G}_t\}_{t \in [t_0, t_1]}$ such that 3.12 and 3.11 hold. Let*

$$V(t, x) = (\mathcal{T}_{tt_1} \psi)(x).$$

If $V \in C(Q)$ then it is a viscosity solution of 3.14.

Proof. Let us prove the subsolution condition. Let $w \in \mathcal{D}$ and (t, x) maximizer of $V - w$ in $\overline{Q}$ and $V(t, x) = w(t, x)$. Then $V \leq w$ on Q , which implies by 3.7 that we have:

$$(\mathcal{T}_{tr} w(r, \cdot))(x) \geq (\mathcal{T}_{tt_1} \psi)(x) = V(t, x) = w(t, x). \quad (3.17)$$

Now, because of 3.12 and 3.11 we can compute:

$$-\frac{\partial}{\partial t} w(t, x) + (\mathcal{G}_t w(t, \cdot))(x) = -\lim_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{t+h} w(t+h, \cdot))(x) - w(t, x)] \leq 0$$

because of 3.17. The supersolution condition is proven analogously. $\square$

3.2 The Standard Approach

We have talked about *viscosity solutions* for $\mathcal{G}_t$ that are infinitesimal generators of semigroups. The notion of viscosity solution was introduced for partial differential equations, that is when $\mathcal{G}_t$ is a partial differential operator. In this section we will introduce this classical notion and provide some properties, establishing as well the relevant links to what we have already proven.

Let $\mathcal{G}_t$ be a partial differential operator defined as

$$(\mathcal{G}_t \phi)(x) = F(t, x, D\phi(x), D^2\phi(x), \phi(x)), \quad (3.18)$$

where F is taken to be continuous. We assume that $\sigma' = O \subset \mathbb{R}^n$ open and σ to be its closure. Furthermore, we assume that test functions $\mathcal{C}$ and $\mathcal{D}$ are such that

$$C_p(\overline{O}) \cap \mathcal{M}(\overline{O}) \subset \mathcal{C}, \quad C_p^\infty(\overline{Q}) \cap \mathcal{M}(\overline{Q}) \subset \mathcal{D}, \quad (3.19)$$

where $Q = [t_0, t_1] \times O$ as usual. A useful result is the following.

Lemma 3.2.1. *The operator $\mathcal{G}_t$ obeys to the maximum principle if and only if F is elliptic.*

Proof. Recall that F is elliptic if

$$F(t, x, p, A + B, V) \leq F(t, x, p, A, V) \quad (3.20)$$

for all $(t, x, p, V) \in Q \times \mathbb{R}^n \times \mathbb{R}$ and all $A, B \in \text{Sym}(\mathbb{R}^{n \times n})$ with $B \geq 0$.

Let us assume that F is elliptic, then for generic $\phi, \psi \in C^2(O)$ and $\bar{x} \in \arg \max\{(\phi - \psi)(x) | x \in \overline{O}\} \cap O$ with $\phi(\bar{x}) = \psi(\bar{x})$ we have that $D\phi(\bar{x}) = D\psi(\bar{x})$ and $D^2\phi(\bar{x}) \leq D^2\psi(\bar{x})$, therefore

$$F(t, \bar{x}, D\phi(\bar{x}), D^2\phi(\bar{x}), \phi(\bar{x})) \geq F(t, \bar{x}, D\psi(\bar{x}), D^2\psi(\bar{x}), \psi(\bar{x})),$$

just by using 3.20 with $B = D^2\psi(\bar{x}) - D^2\phi(\bar{x}) \geq 0$. If F is not elliptic for some $(t, \bar{x}, p, A, V)$ then the functions

$$\psi(x) = V + p \cdot (x - \bar{x}) + \frac{1}{2}(A + B)(x - \bar{x}) \cdot (x - \bar{x}),$$

$$\phi(x) = V + p \cdot (x - \bar{x}) + \frac{1}{2}A(x - \bar{x}) \cdot (x - \bar{x}),$$

for $x \in \bar{O}$ are such that $\bar{x}$ maximizes the distance between them, they coincide over it, but the value of F in $\bar{x}$ and ϕ, ψ does not satisfy the maximum principle, since 3.20 does not hold. $\square$

In this context we give the following definition.

Definition 3.2.1. *Given a function $W \in C(\bar{O})$ we define the equation*

$$-\frac{\partial}{\partial t}W(t, x) + F(t, x, D_x W(t, x), D_x^2 W(t, x), W(t, x)) = 0 \quad (3.21)$$

(a) *W is a viscosity subsolution of 3.21 in Q if for each $w \in C^\infty(Q)$ then*

$$-\frac{\partial}{\partial t}w(t, x) + F(t, x, D_x w(t, x), D_x^2 w(t, x), w(t, x)) \leq 0 \quad (3.22)$$

at every $(t, x) \in Q$ which locally maximizes $W - w$ yielding 0.

(b) *W is a viscosity supersolution of 3.21 in Q if for each $w \in C^\infty(Q)$ then*

$$-\frac{\partial}{\partial t}w(t, x) + F(t, x, D_x w(t, x), D_x^2 w(t, x), w(t, x)) \geq 0 \quad (3.23)$$

at every $(t, x) \in Q$ which locally minimizes $W - w$ yielding 0.

(c) *W is a viscosity solution if it is both a viscosity subsolution and supersolution.*

We will prove that definitions 3.1.1 and 3.2.1 are equivalent for partial differential operators of the form 3.18, provided that the function has sub-polynomial growth and is measurable.

A handy way of verifying both the definitions is the following.

Lemma 3.2.2. *If the conditions in the definitions 3.1.1, 3.2.1 is satisfied only at strict extrema then it is satisfied at every extrema.*

Proof. We prove it for the minimum of definition 3.2.1, the other assertions are proved similarly. Let (t, x) be a non strict minimum of $W - w$ such that $W(t, x) = w(t, x)$. Let $\epsilon > 0$ and define $w^\epsilon = w + \epsilon\xi$ where

$$\xi(s, y) = e^{-(|s-t|^2 + |y-x|^2)} - 1$$

defined over $\overline{Q}$. Since $\mathcal{D}$ is a vector space and $\epsilon\xi \in \mathcal{D}$ by 3.19, then $w^\epsilon \in \mathcal{D}$. Then for every $\epsilon > 0$ the function $W - w^\epsilon$ has a strict minimum at (t, x) with $W(t, x) = w^\epsilon(t, x)$ which implies that

$$-\frac{\partial}{\partial t}w^\epsilon(t, x) + F(t, x, D_x w^\epsilon(t, x), D_x^2 w^\epsilon(t, x), w^\epsilon(t, x)) \leq 0.$$

By continuity of F and smoothness of w^ϵ we let $\epsilon \rightarrow 0$ and get the wanted inequality. $\square$

Finally, we prove the equivalence of these definitions.

Theorem 3.2.3. *Let all the previous assumptions and $W \in C_p(\overline{Q}) \cap \mathcal{M}(\overline{Q})$ and $\mathcal{D} \subset C^{1,2}(Q)$. Then W is a subsolution (supersolution) of 3.21 as defined by 3.1.1 if and only if it is a subsolution (supersolution) as defined by 3.2.1.*

Proof. Let W be a supersolution as in 3.1.1. We just prove the supersolution property, since if we have W subsolution we can see it as a supersolution. Let W a subsolution and (t, x) a strict maximum of $W - w$ and $W(t, x) = w(t, x)$, then we can find an open set A such that $w > W$ over $A \cap Q$. By sub-polynomial growth there exists a constant K such that

$$|W(s, y)| \leq K(1 + |y|^{2m}), \quad \forall (s, y) \in \overline{Q}$$

for some $m > 0$. Then for some ξ given by Urysohn's lemma we can define

$$\overline{w}(s, y) = \xi(s, y)w(s, y) - (1 - \xi(s, y))K(1 + |y|^{2m}),$$

and get a minimum for $W - \overline{w}$ in (t, x) . Then we are in the supersolution case. Let $w \in C^\infty(Q)$ and (t, x) a local minimum of $W - w$ where it is null. There exists an open subset $\mathcal{N}$ of $\mathbb{R}^n$ such that $W \geq w$ over $\mathcal{N} \cap \overline{Q}$, and by Urysohn's lemma there exists a function $\xi \in C^\infty(\overline{Q})$ taking values in $[0, 1]$, constantly 1 in a neighborhood $\tilde{\mathcal{N}}$ of (t, x) and zero outside $\mathcal{N}$. Since W is bounded below we define

$$\tilde{w}(s, y) = \xi(s, y)w(s, y) - (1 - \xi(s, y))K, \quad (3.24)$$

where $-K$ is the lower bound of W . Then, by hypotheses on W , w , and ξ we have

$$W = \xi W + (1 - \xi)W \geq \xi w + (1 - \xi)W \geq \xi w - (1 - \xi)K = \overline{w}.$$

Furthermore, $\overline{w} \in \mathcal{D}$ since 3.19 holds, (t, x) is a minimum of $W - \overline{w}$, and $W(t, x) = \overline{w}(t, x)$ then

$$-\frac{\partial}{\partial t}\overline{w}(t, x) + F(t, x, D_y \overline{w}(t, x), D_y^2 \overline{w}(t, x), \overline{w}(t, x)) \geq 0,$$

but $\overline{w}$ and w have same derivative in (t, x) , since ξ is constantly unitary in a neighborhood of it. Then we get

$$-\frac{\partial}{\partial t}w(t, x) + F(t, x, D_y w(t, x), D_y^2 w(t, x), w(t, x)) \geq 0.$$

Let us suppose that W is a subsolution for 3.2.1. Let $w \in \mathcal{D}$ and (t, x) a strict maximum of $W - w$ such that $(W - w)(t, x) = 0$. Since $\mathcal{D} \subset C^{1,2}(Q)$ we can find an open set Q^* over which w is $C^{1,2}$ and then extend it to be null over $\mathbb{R}^{n+1} \setminus Q^*$. Let us define a sequence of mollifications of w defined as w^n such that $w^n, w_t^n, w_{x_i}^n, w_{x_i x_j}^n$ converge to their respective of w . Since the convergence of w^n to w is uniform there must be a subsequence (t_n, x_n) converging to (t, x) , such that $W - w^n$ has a maximum. If we set $\bar{w}^n = w^n - w^n(t_n, x_n) + W(t_n, x_n)$ we still have a maximum in (t_n, x_n) for $W - \bar{w}^n$ and $W(t_n, x_n) = \bar{w}^n(t_n, x_n)$. Then

$$-\frac{\partial}{\partial t} \bar{w}^n(t_n, x_n) + F(t_n, x_n, D_x \bar{w}^n(t_n, x_n), D_x^2 \bar{w}^n(t_n, x_n), \bar{w}^n(t_n, x_n)) \leq 0.$$

Because of continuity of F and smoothness of $\bar{w}^n$ we get

$$-\frac{\partial}{\partial t} w(t, x) + F(t, x, D_y w(t, x), D_y^2 w(t, x), w(t, x)) \leq 0.$$

□

3.3 Value Function as Viscosity Solution

In the first section we proved that the value function of a dynamic programming problem is the classical solution of the dynamic programming equation, under differentiability assumptions. We now prove that it is a viscosity solution of the dynamic programming equation given two sets of assumptions.

Theorem 3.3.1. *Let U be a bounded space of control and $f \in C(\bar{Q} \times U)$ such that $|f(t, x, v)| \leq K(1 + |x|)$. Then for every $w \in C^1(Q) \cap \mathcal{M}(\bar{Q})$*

$$\lim_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{t+h} w(t + h, \cdot))(x) - w(t, x)] = \frac{\partial}{\partial t} w(t, x) - H(t, x, D_x w(t, x)), \quad \forall (t, x) \in \bar{Q}. \quad (3.25)$$

Note that equation 3.25 implies that V is a viscosity solution of 1.14 via Theorem 3.1.2.

Proof. Let $(t, x) \in Q$, $v \in U$ and $w \in C^1(Q) \cap \mathcal{M}(\bar{Q})$. We prove that

$$\limsup_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{t+h} w(t + h, \cdot))(x) - w(t, x)] \leq \frac{\partial}{\partial t} w(t, x) - H(t, x, D_x w(t, x)), \quad (3.26)$$

and

$$\liminf_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{t+h} w(t + h, \cdot))(x) - w(t, x)] \geq \frac{\partial}{\partial t} w(t, x) - H(t, x, D_x w(t, x)). \quad (3.27)$$

Equation 3.26 can be proven without assuming boundedness of the control space U and sub-linearity of the flow f . Indeed, there exists an admissible control $u(\cdot) \in \mathcal{U}(t, x)$ such that $\lim_{s \rightarrow t} u(s) = v$. Let $x(\cdot)$ be the state solution of the Cauchy problem associated to the control $u(\cdot)$ such that $x(t) = x$. Let also τ the exit time of $(s, x(s))$ from Q . If we take a sufficiently small $h > 0$ we can compute

$$\begin{aligned} \frac{1}{h} [(\mathcal{T}_{tt+h} w(t+h, \cdot))(x) - w(t, x)] &\leq \frac{1}{h} \int_t^{t+h} L(s, x(s), u(s)) ds \\ &\quad + \frac{1}{h} [w(t+h, x(t+h)) - w(t, x)] \\ &= \frac{1}{h} \int_t^{t+h} \left[L(s, x(s), u(s)) \right. \\ &\quad \left. + \frac{\partial}{\partial t} w(s, x(s)) + f(s, x(s), u(s)) \cdot D_x w(s, x(s)) \right] ds. \end{aligned}$$

Then, 3.26 holds true.

Let us prove 3.27. Using $|f(t, x, v)| \leq K(1 + |x|)$ we get

$$\begin{aligned} |x(r) - x| &\leq \int_r^t |f(s, x(s), u(s))| ds \\ &\leq K(r - t) + K \int_t^r |x|(s) ds \\ &\leq K(r - t)(1 + |x|) + K \int_t^r |x(s) - x| ds, \end{aligned}$$

which implies by Gronwall's inequality

$$|x(r) - x| \leq (1 + |x|) e^{K(t-r)-1}, \quad \forall r \geq t.$$

Then for suitably small r that $\tau \geq r \wedge t_1$ for all $u(\cdot) \in \mathcal{U}(t, x)$. Then for n sufficiently large by definition of $\mathcal{T}_{tr}$ we can find a control $u^n(\cdot)$ such that

$$\left(\mathcal{T}_{tt+\frac{1}{n}} w \left(t + \frac{1}{n} \right) \right) (x) \geq \int_t^{t+1/n} L(s, x(s), u(s)) ds + w \left(t + \frac{1}{n}, x^n \left(t + \frac{1}{n} \right) \right) - \frac{1}{n^2},$$

which implies that

$$\begin{aligned} n \left[\left(\mathcal{T}_{tt+\frac{1}{n}} w \left(t + \frac{1}{n} \right) \right) (x) - w(t, x) \right] &\geq \frac{\partial}{\partial t} w(t, x) + n \int_t^{t+1/n} L(s, x(s), u^n(s)) ds \\ &\quad + n \int_t^{t+1/n} f(t, x, u^n(s)) ds \cdot D_x w(t, x) + e(n), \end{aligned} \tag{3.28}$$

where

$$\begin{aligned}
e(n) = & -\frac{1}{n} + n \int_t^{t+1/n} \frac{\partial}{\partial t} (w(s, x^n(s)) - w(t, x)) ds \\
& + n \int_t^{t+1/n} L((s, x^n(s), u^n(s))) - L(t, x, u^n(s)) ds \\
& + n \int_t^{t+1/n} f(s, x^n(s), u^n(s)) \cdot D_x w(s, x^n(s)) - f(t, x, u^n(s)) \cdot D_x w(t, x) ds.
\end{aligned} \tag{3.29}$$

Because of the continuity of L , of f and smoothness of w we have that $\lim_{n \rightarrow +\infty} e(n) = 0$. If we prove that

$$\begin{aligned}
n \int_t^{t+1/n} L(s, x(s), u^n(s)) ds + n \int_t^{t+1/n} f(t, x, u^n(s)) ds \cdot D_x w(t, x) \\
\leq \inf_{v \in U} \{f(t, x, v) \cdot p + L(t, x, v)\},
\end{aligned} \tag{3.30}$$

then from 3.28 we get

$$\liminf_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{t+h} w(t+h))(x) - w(t, x)] \geq \frac{\partial}{\partial t} w(t, x) - H(t, x, D_x w(t, x)).$$

□

We state a general result for later use.

Corollary 3.3.2. *Let U be a bounded space of control and $f \in C(\overline{Q} \times U)$ such that $|f(t, x, v)| \leq K(1 + |x|)$, then the value function is a solution to the dynamic programming equation 1.14.*

It is important to notice that the previous result holds under quite stringent hypotheses. We both require the space control to be bounded and the flow f to have sublinear growth in its state variable. We can relax these assumptions and get a similar result, but we need to require the existence of an optimal control.

Theorem 3.3.3. *If for each $(t, x) \in Q$ there exists a $u^* \in \mathcal{U}(t, x)$ be an optimal control, then a continuous value function is a viscosity solution of its dynamic programming equation.*

Proof. Let us drop the *. As we did before in Theorem 3.3.1, for every $w \in C^\infty(Q)$ we have

$$\limsup_{h \rightarrow 0} \frac{1}{h} [(\mathcal{T}_{t+h} w(t+h, \cdot))(x) - w(t, x)] \leq \frac{\partial}{\partial t} w(t, x) - H(t, x, D_x w(t, x)).$$

If $V - w$ has a maximum at (t, x) and $V(t, x) = w(t, x)$ then

$$\limsup_{h \rightarrow 0} \frac{1}{h} \left[\underbrace{\int_t^{t+h} L(s, x(s), u(s)) ds}_{=V(t,x)-w(t,x)=0} + w(t+h, x) - V(t+h, x) \right] = 0,$$

which proves subsolution. Let the subsolution hypotheses hold, then from the dynamic programming principle we get

$$\begin{aligned} 0 &\geq \int_t^{t+h} L(s, x(s), u(s)) ds + w(t+h, x(t+h)) - w(t, x) \\ &\geq \int_t^{t+h} L(s, x(s), u(s)) + \frac{\partial}{\partial t} w(s, x(s)) + f(s, x(s), u(s)) \cdot D_x w(s, x(s)) ds \\ &\geq \int_t^{t+h} L(s, x(s), u(s)) - H(s, x(s), D_x w(s, x(s))) ds. \end{aligned}$$

Finally, if we let $h \rightarrow 0$ we get

$$-\frac{\partial}{\partial t} w(t, x) + H(t, x, D_x w(t, x)) \geq 0.$$

□

We now turn our attention to uniqueness property of viscosity solution to the dynamic programming equation. Let us consider the dynamic programming equation in a more general form

$$-\frac{\partial}{\partial t} V(t, x) + H(t, x, D_x V(t, x)) = 0, (t, x) \in Q. \quad (3.31)$$

We will show that under technical assumptions on H , which are satisfied by the Hamiltonian, the supremum norm of the difference between two viscosity solution can be computed only on the frontier. Therefore, if we impose boundary conditions we get uniqueness of the viscosity solution. The assumptions on H are

1. There exists $K \in \mathbb{R}$ and $h \in C([0, \infty))$ with $h(0) = 0$ such that $\forall (t, x), (s, y) \in \overline{Q}$ and $\forall p, p' \in \mathbb{R}^n$ we have

$$H(t, x, p) - H(s, y, p') \leq h(|t - s| + |x - y|) + h(|t - s|)|p| + K|x - y||p| + K|p - p'|.$$

2. There exists $K' \in \mathbb{R}$ such that

$$|H_t| + |H_x| \leq K'(1 + |p|),$$

and

$$|H_p| \leq K'.$$

Under these hypotheses we can state the following theorem.

Theorem 3.3.4. *Let W and V viscosity subsolution and supersolution of 3.31 in Q , respectively. If Q is unbounded we assume W, V to be bounded and uniformly continuous on its closure. Then*

$$\sup_{\overline{Q}}[W - V] = \sup_{\partial^* Q}[W - V].$$

This Theorem implies that if V and W are two viscosity solution of 3.31 satisfying the boundary conditions

$$\begin{aligned} V(t, x) &= g(t, x), \forall (t, x) \in [T_0, t_1] \times O, \\ V(t_1, x) &= \psi(x), x \in \overline{O}, \end{aligned} \tag{3.32}$$

then, if the assumption of Theorem 3.3.4 are satisfied, we get

$$\|W - V\|_\infty = \sup_{\overline{Q}}[W - V] = \sup_{\partial^* Q}[W - V] = 0.$$

The proof of Theorem 3.3.4 is rather long; we first prove the $\overline{Q}$ bounded case and then the unbounded one.

Proof. Let Q be bounded. For $\epsilon, \delta, \beta > 0$ define

$$\Phi(t, x, s, y) = W(t, x) - V(s, y) - \frac{1}{2\epsilon}|x - y|^2 - \frac{1}{2\delta}|t - s|^2 + \beta(s - t_1).$$

Since $\overline{Q}$ is bounded the function Φ achieves its supremum at $(t', x'), (s', y') \in \overline{Q} \times \overline{Q}$. We will essentially prove that Φ divides $W - V$ from its supremum over the boundary in an inequality chain.

1. Let $\rho > 0$ and define

$$D_\rho = \{(t, x), (s, y) \in \overline{Q} \times \overline{Q} \mid |t - s|^2 + |x - y|^2 \leq \rho\},$$

and

$$m_W(\rho) = 2 \{|W(t, x) - W(s, y)| : ((t, x), (s, y)) \in D_\rho\},$$

$$m_V(\rho) = 2 \{|V(t, x) - V(s, y)| : ((t, x), (s, y)) \in D_\rho\},$$

and

$$K_1 = \sup\{m_w(\rho) : \rho \geq 0\}.$$

Since $\overline{Q}$ is compact and W, V are continuous they are uniformly continuous, therefore $m_W, m_V \in C([0, \infty))$. Clearly $m_W(0) = m_V(0) = 0$. We now claim two estimates on $|x - y|$ and $|t - s|$:

$$\begin{aligned} |t' - s'| &\leq \sqrt{K_1 \delta} \\ |x' - y'| &\leq \sqrt{\epsilon m_W(K_1[\epsilon + \delta])}. \end{aligned}$$

From the maximum in $((t', x'), (s', y'))$ we have

$$\Phi(t', x', s', y') \leq \Phi(s', y', s', y'),$$

implies

$$\frac{1}{\epsilon}|x' - y'|^2 + \frac{1}{\delta}|t' - s'|^2 \leq 2(W(t', x') - W(s', y')) \leq m_W(|t' - s'| + |x' - y'|).$$

Then we have

$$\frac{1}{\delta}|t' - s'|^2 \leq K_1, \quad \frac{1}{\epsilon}|x' - y'|^2 \leq K_1,$$

which also imply

$$\frac{1}{\epsilon}|x' - y'|^2 \leq m_w(K_1[\epsilon + \delta]).$$

2. We now show that if $(t', x') \in \partial^* Q$ or $(s', y') \in \partial^* Q$ we have

$$\Phi(t', x', s', y') \leq \frac{1}{2}m_V(K_1(\epsilon + \delta)) + \sup_{\partial^* Q}[W - V], \quad (3.33)$$

or

$$\Phi(t', x', s', y') \leq \frac{1}{2}m_W(K_1(\epsilon + \delta)) + \sup_{\partial^* Q}[W - V], \quad (3.34)$$

respectively. Indeed, in the first case, we have

$$\begin{aligned} \Phi(t', x', s', y') &\leq W(t', x') - W(s', y') \\ &\leq V(t', x') - V(s', y') + \sup_{\partial^* Q}[W - V] \\ &\leq \frac{1}{2}m_W(|t' - s'|^2 + |x' - y'|^2) + \sup_{\partial^* Q}[W - V]. \end{aligned}$$

Same reasoning for the latter.

3. We now construct an estimate over the parameter β if $(t', x'), (s', y') \in Q$.
Let us consider the test function

$$w(t, x) = \frac{1}{2\delta}|t - s'|^2 + \frac{1}{2\epsilon}|x - y'|^2,$$

then (t', x') is where $W - w$ is maximized and $W(t', x') = w(t', x')$, then

$$-\underbrace{\frac{1}{\delta}(t' - s')}_{\stackrel{def}{=} q_\delta} + H(t', x', \underbrace{\frac{1}{\epsilon}(x' - y')}_{\stackrel{def}{=} p_\epsilon}) \leq 0. \quad (3.35)$$

Analogously, defining

$$w(s, y) = -\frac{1}{2\delta}|t' - s|^2 - \frac{1}{2\epsilon}|x' - y|^2 + \beta(s - t_1),$$

we get:

$$-\beta - q_\delta + H(s', y', p_\epsilon) \geq 0. \quad (3.36)$$

Combining 3.35 and 3.36 we get

$$\begin{aligned} \beta &\leq H(s', y', p_\epsilon) - H(t', x', p_\epsilon) \\ &\leq h(|t' - s'| + |x' - y'|) + h(|t' - s'|)|p_\epsilon| + K|x' - y'||p_\epsilon| \\ &\leq h(\sqrt{K_1\epsilon} + \sqrt{K_1\delta}) + h(\sqrt{K_1\delta})\frac{1}{\epsilon}\sqrt{K_1\epsilon} + K\frac{|x' - y'|^2}{\epsilon} \\ &\leq h(\sqrt{K_1\epsilon} + \sqrt{K_1\delta}) + h(\sqrt{K_1\delta})\sqrt{\frac{K_1\epsilon}{\epsilon}} + Km_W(K_1[\epsilon + \delta]). \end{aligned}$$

4. We denote

$$k(\epsilon, \delta) = h(\sqrt{K_1\epsilon} + \sqrt{K_1\delta}) + h(\sqrt{K_1\delta})\sqrt{\frac{K_1\epsilon}{\epsilon}} + Km_W(K_1[\epsilon + \delta]),$$

then

$$\lim_{\epsilon \rightarrow 0} \lim_{\delta \rightarrow 0} k(\epsilon, \delta) = 0.$$

Then for suitably small ϵ and δ we have

$$k(\epsilon, \delta) < \beta,$$

which implies that either (t', x') or (s', y') is in ∂^*Q . Then by 3.33 or 3.34 we have

$$\limsup_{\epsilon \rightarrow 0} \limsup_{\delta \rightarrow 0} \Phi(t', x', s', y') \leq \sup_{\partial^*Q} [W - V], \quad (3.37)$$

for every $\beta > 0$. Finally, we have

$$W(t, x) - V(s, y) + \beta(t - t_1) = \Phi(t, x, t, x) \leq \Phi(t', x', s', y'),$$

then by 3.37 we get

$$W(t, x) - V(s, y) = \lim_{\beta \rightarrow 0} W(t, x) - V(s, y) + \beta(t - t_1) \leq \lim_{\beta \rightarrow 0} \Phi(t', x', s', y') \leq \sup_{\partial^*Q} [W - V].$$

Let us suppose that Q is unbounded, and V, W are bounded and uniformly continuous. We keep the same notation as before. For every $\gamma > 0$ there exist $(t_\gamma, x_\gamma), (s_\gamma, y_\gamma) \in \overline{Q}$ such that

$$\Phi(t_\gamma, x_\gamma, s_\gamma, y_\gamma) \geq \sup_{\overline{Q} \times \overline{Q}} \Phi - \gamma, \quad (3.38)$$

since W and V are bounded. We therefore define

$$\Phi_\gamma(t, x, s, y) = \Phi(t, x, s, y) - \frac{\gamma}{2} [|t - t_\gamma|^2 + |s - s_\gamma|^2 + |x - x_\gamma|^2 + |y - y_\gamma|^2],$$

for $(t, x), (s, y) \in \overline{Q}$.

We now follow a similar procedure to the one used in the bounded case.

- 1') Let the moduli of continuity m_W, m_V and the bound K_1 as before. Since we assume the value functions to be bounded and absolutely continuous we have, as before, that

$$m_W, m_V \in C([0, \infty)), m_W(0) = m_V(0) = 0, K_1 = \sup\{m_W(\rho) : \rho \geq 0\} < +\infty.$$

Let us note that for $(t, x), (s, y) \in \overline{Q} \times \overline{Q}$ such that

$$|t - t_\gamma|^2 + |s - s_\gamma|^2 + |x - x_\gamma|^2 + |y - y_\gamma|^2 > 2,$$

then

$$\Phi_\gamma(t, x, s, y) \leq \Phi(t, x, s, y) - \gamma \leq \Phi(t_\gamma, x_\gamma, s_\gamma, y_\gamma) = \Phi_\gamma(t_\gamma, x_\gamma, s_\gamma, y_\gamma).$$

The supremum of Φ_γ is achieved since Φ_γ is bounded by $\Phi(t_\gamma, x_\gamma, s_\gamma, y_\gamma)$. It is achieved at $(t', x'), (s', y')$. Then

$$|t - t_\gamma|^2 + |s - s_\gamma|^2 + |x - x_\gamma|^2 + |y - y_\gamma|^2 \leq 2.$$

We now claim two estimates on the distance between the time variables and space variables. For $\gamma, \delta, \epsilon \leq \frac{1}{2}$ we have

$$|t' - s'| \leq \sqrt{2(K_1 + 1)\delta}, \quad |x' - y'| \leq \sqrt{2\epsilon(2\gamma + m_W(2\epsilon(K_1 + 1)))}. \quad (3.39)$$

Indeed, $\Phi_\gamma(t', x', s', y') \geq \Phi_\gamma(s', x', s', y')$ implies

$$\begin{aligned} \frac{1}{2\delta}|t' - s'|^2 &\leq W(t', x') - W(s', x') + \frac{\gamma}{2}[|s' - t_\gamma|^2 - |t' - t_\gamma|^2] \\ &\leq \frac{1}{2}K_1 + \gamma|t' - s'|^2 + \frac{\gamma}{2}|t' - t_\gamma|^2 \\ &\leq \frac{1}{2}K_1 + \gamma|t' - s'|^2 + \gamma, \end{aligned}$$

which implies, recalling the bound on δ and γ , that

$$\left(\frac{1}{4\delta}\right)|t' - s'|^2 \leq \left(\frac{1-\delta}{2\delta}\right)|t' - s'|^2 \leq \left(\frac{1}{2\delta} - \gamma\right)|t' - s'|^2 \leq \frac{1}{2}K_1 + \frac{1}{2}.$$

From $\Phi_\gamma(t', x', s', y') \geq \Phi_\gamma(t', y', s', y')$ we get

$$\begin{aligned} \frac{1}{2\epsilon}|x' - y'|^2 &\leq W(t', x') - W(t', y') + \frac{\gamma}{2}[|y' - x_\gamma|^2 - |x' - x_\gamma|^2] \\ &\leq \frac{1}{2}m_W(|x' - y'|^2) + \gamma|x' - y'|^2 + \frac{\gamma}{2}|x' - x_\gamma|^2 \\ &\leq \frac{1}{2}m_W(|x' - y'|^2) + \gamma|x' - y'|^2 + \gamma, \end{aligned}$$

which implies

$$\frac{1}{4\epsilon}|x' - y'|^2 \leq \frac{1}{2}m_W(|x' - y'|^2) + \gamma \Rightarrow |x' - y'|^2 \leq 2(K_1 + 1)\epsilon.$$

Therefore, we get

$$\frac{1}{4\epsilon}|x' - y'|^2 \leq \left(\frac{1}{2\epsilon} - \gamma\right)|x' - y'|^2 \leq \frac{1}{2}m_W(2(K_1 + 1)\epsilon) + \gamma,$$

hence the second equation in 3.39.

2') Replicating the same argument of the bounded case we get if $(t', x') \in \partial^* Q$ then

$$\Phi_\gamma(t', x', s', y') \leq \frac{1}{2}m_V (2(K_1 + 1)(\epsilon + \delta)) + \sup_{\partial^* Q} [W - V],$$

and if $(s', y') \in \partial^* Q$ then

$$\Phi_\gamma(t', x', s', y') \leq \frac{1}{2}m_W (2(K_1 + 1)(\epsilon + \delta)) + \sup_{\partial^* Q} [W - V].$$

3') Finally, if we find a bound over β for $(t', x'), (s', y') \in Q$ we then prove that this bound is not satisfied and then we can invoke point 2' and get

$$\limsup_{\epsilon \rightarrow 0} \limsup_{\delta \rightarrow 0} \limsup_{\gamma \rightarrow 0} \Phi_\gamma(t', x', s', y') \leq \sup_{\partial^* Q} [W - V],$$

which implies, as before, that

$$W(t, x) - V(s, y) \leq \sup_{\partial^* Q} [W - V].$$

Let $(t', x'), (s', y') \in Q$ and the test function

$$\bar{w}(t, x) = \frac{1}{2\delta}|t - s'|^2 + \frac{1}{2\epsilon}|x - y'|^2 + \frac{\gamma}{2} (|t - t_\gamma|^2 + |x - x_\gamma|^2),$$

defined over Q . Then (t', x') is where $W - \bar{w}$ is null and maximized. Therefore

$$-\underbrace{\frac{1}{\delta}(t' - s')}_{\stackrel{def}{=} q_\delta} - \underbrace{\gamma(t' - t_\gamma)}_{\stackrel{def}{=} q_\gamma} + H(t', x', p_\epsilon + p_\gamma) \leq 0, \quad (3.40)$$

where

$$p_\epsilon = \frac{1}{\epsilon}(x' - y'), \quad p_\gamma = \gamma(x' - x_\gamma).$$

By the same reasoning we define the test function

$$w^*(s, y) = -\frac{1}{2\delta}|t' - s|^2 - \frac{1}{2\epsilon}|x' - y|^2 + \frac{\gamma}{2} (|s - s_\gamma|^2 + |y - y_\gamma|^2) + \beta(s - t_1),$$

from which we get

$$-\beta - q_\delta - \underbrace{\gamma(s' - s_\gamma)}_{\stackrel{def}{=} \bar{q}_\gamma} + H(s', y', p_\epsilon + \bar{p}_\gamma) \geq 0, \quad (3.41)$$

where $\bar{p}_\gamma = \gamma(y_\gamma - y')$.

We can find the upper bound to β using the estimates in 3.39 and equations 3.41 and 3.40:

$$\begin{aligned}
\beta &\leq H(s', y', p_\epsilon + \bar{p}_\gamma) - H(t', x', p_\epsilon + p_\gamma) + q_\gamma - \bar{q}_\gamma \\
&\leq h(|t' - s'| + |x' - y'|) + h(|t' - s'|) [|p_\epsilon + |\bar{p}_\gamma||] \\
&\quad + K|x' - y'| [|p_\epsilon + |\bar{p}_\gamma||] + K|\bar{p}_\gamma - p_\gamma| + q_\gamma - \bar{q}_\gamma \\
&\leq h(\sqrt{2(K_1 + 1)}\delta + \sqrt{2(K_1 + 1)}\epsilon) + h(\sqrt{2(K_1 + 1)}\delta) \left[\sqrt{\frac{|2(K_1 + 1)|}{\epsilon}} + 2\gamma \right] \\
&\quad + K2 \left[2\gamma + m_W(2(K_1 + 1))\epsilon + \sqrt{2(K_1 + 1)}\epsilon\delta \right] + 2(K + 1)\gamma.
\end{aligned}$$

□

3.3.1 Continuity Properties of the Value Function

We now investigate continuity properties of the value function. In particular, we consider two different sets of assumptions regarding continuity: Lipschitz continuity and Uniform continuity. Let us consider f, L, ψ as usual, recalling that there exists $K_\rho > 0$ such that

$$|f(t, x, v) - f(t, y, v)| \leq K_\rho |x - y|, \quad (3.42)$$

for all $|v| \leq \rho$. Note that these are minimal assumption to start working with the optimal control problem.

Theorem 3.3.5. *Let a bounded control space U , $Q = [t_0, t_1] \times \mathbb{R}^n$. Assume that f, L, ψ are bounded, f satisfies 3.42 and L, ψ uniformly continuous. Then the value function V is bounded and uniformly continuous.*

Proof. V is bounded. Indeed

$$V(t, x) = \inf_{u \in \mathcal{U}} \int_t^{t_1} L(s, x(s), u(s)) ds + \psi(t_1, x(t_1)) \leq (t_1 - t)K_L + K_\psi,$$

where K_g is the bound over the function g .

V is uniformly continuous. Standard arguments on solutions of ODE that we have already used many times (integral form of the solution, bound on the flow and Gronwall) yield

$$\Lambda(s) \stackrel{\text{def}}{=} |x(s) - y(s)| \leq |x - y| e^{K(s-t)}, \quad (3.43)$$

where $x(s), y(s)$ are solution to the Cauchy problem

$$\frac{d}{ds} w(s) = f(s, w(s), u(s)), \quad w(t) = w.$$

Then

$$\begin{aligned}
|J(t, x, u) - J(t, y, u)| &\leq \int_t^{t_1} |L(s, x(s), u(s)) - L(s, y(s), u(s))| ds \\
&\quad + |\psi(x(t_1)) - \psi(y(t_1))| \\
&\leq \int_t^{t_1} m_L(\Lambda(s)) ds + m_\psi(\Lambda(t_1)) \\
&\quad + (t_1 - t_0)m_L(|x - y|e^{K(t_1-t_0)}) + m_\psi(|x - y|e^{K(t_1-t_0)}),
\end{aligned}$$

where m_g is the modulus of continuity of the function g . Since both L and ψ are uniformly continuous $m_L, m_\psi \in C([0, \infty))$ and they are null at the origin. Then for a fixed t we have the bound

$$|V(t, x) - V(t, y)| \leq (t_1 - t_0)m_L(|x - y|e^{K(t_1-t_0)}) + m_\psi(|x - y|e^{K(t_1-t_0)}). \quad (3.44)$$

Similarly, if we fix x we get

$$|V(t, x) - V(s, x)| \leq K_1|t - s| + (t_1 - t_0)m_L(|t - s|K_2e^{K(t_1-t_0)}) + m_\psi(|t - s|K_2e^{K(t_1-t_0)}), \quad (3.45)$$

where $K_1 = \sup |L|$ and K_2 is such that $|x(s) - x| \leq K_2|t - s|$, which exists because f is bounded. Then by splitting $|V(t, x) - V(s, y)|$ we find a modulus of continuity for it, by summing 3.45 and 3.44. $\square$

From the proof it is clear that by taking L and ψ Lipschitz continuous we get V Lipschitz continuous, since we can use

$$m_L(\rho) = K_L\rho, \quad m_\psi(\rho) = K_\psi\rho,$$

for some constants $K_L, K_\psi > 0$.

This Theorem allows us to conclude the following result.

Corollary 3.3.6. *Under the previous assumptions, the value function is the unique viscosity solution of the dynamic programming equation with fixed terminal conditions 3.32.*

Proof. It is a solution because of 3.3.2. We prove uniqueness via Theorem 3.3.4. Since U is bounded the Hamiltonian H satisfies 3.3. Indeed, the bound on the derivative is obvious. The upper bound function h can be found as follows:

$$\begin{aligned}
H(t, x, p) - H(s, y, p') &\leq \sup_{v \in U} \{ |L(t, x, v) - L(s, y, v)| + f(t, x, v) \cdot p - f(s, y, v) \cdot p' \} \\
&\leq \sup_{v \in U} \{ |L(t, x, v) - L(s, y, v)| \} + \sup_{v \in U} |f(s, y, v)| |p - p'| \\
&\quad + \sup_{v \in U} |f(t, x, v) - f(s, y, v)| |p|,
\end{aligned}$$

but the uniform continuity of L and Lipschitz continuity of f yield 3.3. Then by Theorem 3.3.4 we get the thesis. $\square$

3.4 Pontryagin's Principle for Viscosity Solutions

In section 1.3 we stated the Pontryagin's principle for continuously differentiable value functions. It is now clear that value functions may easily fail to be even differentiable, and thus the necessity of a weaker notion of solution. We have introduced the concept of viscosity solution, which turns out to be a key concept to extend Pontryagin's principle in non-differentiable settings. Recall that Pontryagin states the relation between a so-called adjoint variable P and the state variable via a system of differential equations. Since the Hamiltonian may fail to be differentiable, we need to carefully extend the possible values of the adjoint variable for those instances. Let us define two new key concepts that incorporate the possible value that the incremental limit may have in non-smooth points.

Definition 3.4.1. *Let $W \in C(\overline{Q})$ and $(t, x) \in Q$. We define:*

1. *The set of superdifferentials of W at (t, x) as the collection of all $(q, p) \in \mathbb{R} \times \mathbb{R}^n$ such that there exists some $w \in C^1(Q)$ for which*

$$(q, p) = \left(\frac{\partial}{\partial t} w(t, x), D_x w(t, x) \right), \quad (3.46)$$

and

$$(t, x) \in \arg \max \{ (W - w)(s, y) \mid (s, y) \in \overline{Q} \}. \quad (3.47)$$

We denote it with $D^+W(t, x)$.

2. *The set of subdifferentials of W at (t, x) as the collection of all $(q, p) \in \mathbb{R} \times \mathbb{R}^n$ such that there exists some $w \in C^1(Q)$ for which*

$$(q, p) = \left(\frac{\partial}{\partial t} w(t, x), D_x w(t, x) \right), \quad (3.48)$$

and

$$(t, x) \in \arg \min \{ (W - w)(s, y) \mid (s, y) \in \overline{Q} \}. \quad (3.49)$$

We denote it with $D^-W(t, x)$.

Note that this definition is consistent with the notion of viscosity solution since if W is a viscosity subsolution in Q if and only if

$$-q + H(t, x, p) \leq 0, \quad \forall (q, p) \in D^+W(t, x), \quad \forall (t, x) \in Q,$$

and similarly for a supersolution.

Example 3.4.1. In subsection 3.0.1 we showed a non-differentiable value function. Recall that the original problem was

$$\inf_{x \in \text{Lip}([0,1];[-1,1])} \int_t^{t_1} 1 + \frac{1}{4}(\dot{x}(s))^2 ds, \quad (3.1)$$

and the resulting value function

$$V(t, x) = \begin{cases} 1 - |x| & |x| \leq t \\ 1 - t & |x| \geq t. \end{cases} \quad (3.2)$$

Let us compute $D^+V(t, x)$ and $D^-V(t, x)$. If (t, x) are not of the form $t = |x|$ the value function is differentiable and the superdifferential and subdifferential coincide with the derivative. Let $a \in [0, 1]$ let us compute $D^+V(a, a)$, we compute all the gradients the function can take, the superdifferential will be their convex. When $x > 0$ at $t = x$, $V(t, x) = 1 - t$. The partial derivatives when approaching from different sides are:

$$\begin{aligned} \frac{\partial V}{\partial t} &= -1 & (\text{approaching from the } t < |x| \text{ side}). \\ \frac{\partial V}{\partial x} &= -1 & (\text{approaching from the } t > |x| \text{ side}). \end{aligned}$$

When $x < 0$, at $t = -x$, $V(t, x) = 1 - t$. The partial derivatives when approaching from different sides are:

$$\begin{aligned} \frac{\partial V}{\partial t} &= -1 & (\text{approaching from the } t < |x| \text{ side}). \\ \frac{\partial V}{\partial x} &= 1 & (\text{approaching from the } t > |x| \text{ side}). \end{aligned}$$

The superdifferential is given by the convex combination of these gradients:

$$D^+V(t, x) = \left\{ \lambda(-1, 0) + (1 - \lambda) \left(0, -\frac{x}{|x|} \right) : \lambda \in [0, 1] \right\}.$$

The notion of superdifferential will take the place of the derivative in Pontryagin's result. We recall the definition of the adjoint variable for a state variable x defined by the flow f , a control u , a terminal condition ψ , a Lagrangian L and a Hamiltonian H :

$$\dot{p}_j^*(s) = - \sum_{i=1}^n \frac{\partial}{\partial x_j} f_i(s, x^*(s), u^*(s)) p_i(s) - \frac{\partial}{\partial x_j} L(s, x^*, u^*), \quad (3.50)$$

And also:

$$p(s) \cdot f(s, x^*(s), u^*(s)) + L(s, x^*(s), u^*(s)) = -H(s, x^*(s), u^*(s), p^*(s)), \quad (3.51)$$

With:

$$p^*(t_1) = D\psi(x^*(t_1)). \quad (3.52)$$

We can now state Pontryagin's maximum principle in this broader context.

Theorem 3.4.1. *Let $u^*(\cdot)$ be an optimal control at (t, x) which is right continuous at each $[t, t_1)$, and $p^*(s)$ defined by 3.50, 3.51 and 3.52. Then for each $s \in [t, t_1)$*

$$\left(H(s, x^*(s), p^*(s)), p^*(s) \right) \in D^+V(s, x^*(s)). \quad (3.53)$$

Proof. By definition of value function we have

$$V(r, y) \leq J(r, y; u^*), \quad \forall (r, y) \in Q,$$

and equality for $(r, y) = (s, x^*(s))$ for each $s \in [t, t_1]$. Hence, if we show that J is continuously differentiable and

$$\frac{\partial}{\partial r} J(s, x^*(s); u^*) = p^*(s), \quad (3.54)$$

and

$$D_y J(s, x^*(s); u^*) = H(s, x^*(s), p^*(s)) \quad (3.55)$$

for $s \in [t, t_1)$, then the thesis holds by definition. We prove 3.54 by direct computation

$$\begin{aligned} \frac{\partial}{\partial x_i} J(s, x^*(s); u^*) &= \sum_{j=1}^n \left\{ \int_s^{t_1} L_{x_j}(r, x^*(r), u^*(r)) \frac{\partial}{\partial x_j} x_i(r, x^*(r)) dr \right. \\ &\quad \left. + \psi_{x_j}(x^*(t_1)) \frac{\partial}{\partial x_i} x_j(t_1, x^*(t_1)) \right\}, \end{aligned}$$

but we have

$$\frac{d}{dr} \left\{ \sum_{j=1}^n \frac{\partial}{\partial x_i} x_j(r, x^*(r)) p_i(r) \right\} = - \sum_{i=1}^n \frac{\partial}{\partial x_j} L(s, x^*(r), u^*(r)) \frac{\partial}{\partial x_i} x_j(r, x^*(r)), \quad (3.56)$$

then by integrating and recalling that

$$\frac{\partial}{\partial x_i} x_j(s, x^*(s)) = \delta_{ij}$$

we get

$$\begin{aligned}
p_i(s) &= \sum_{j=1}^n \frac{\partial}{\partial x_i} x_j(s, x^*(s)) p_j(s) \\
&= \sum_{j=1}^n \frac{\partial}{\partial x_i} x_j(t_1, x^*(t_1)) p_j(t_1) - \int_s^{t_1} \frac{d}{dr} \left\{ \sum_{j=1}^n \frac{\partial}{\partial x_i} x_j(r, x^*(r)) p_j(r) \right\} dr \\
&= \frac{\partial}{\partial x_i} J(s, x^*(s); u^*) \\
&= \frac{\partial}{\partial x_i} V(s, x^*(s)).
\end{aligned}$$

To prove 3.54 we define

$$z(s; r, y) = \frac{\partial}{\partial r} x(s; r, y),$$

which allows us to compute

$$\begin{aligned}
\frac{\partial}{\partial r} J(r, x^*(r); u^*) &= -L(r, x^*(r), u^*(r)) \\
&\quad + \sum_{j=1}^n \left\{ \int_r^{t_1} \frac{\partial}{\partial x_j} L(s, x^*(s), u^*(s)) z_j(s; r, x^*(r)) ds \right. \\
&\quad \left. + \frac{\partial}{\partial x_j} \psi(t_1, x^*(t_1)) z_j(t_1; r, x^*(r)) \right\},
\end{aligned}$$

but

$$\frac{d}{ds} [z(s) \cdot p(s)] = - \sum_{j=1}^n \frac{\partial}{\partial x_j} L(s, x^*(s), u^*(s)) z_j(s),$$

then

$$\begin{aligned}
-f(r, x^*(r), u^*(r)) \cdot p(r) &= z(r) \cdot p(r) \\
&= z(t_1) \cdot p(t_1) - \int_r^{t_1} \frac{d}{ds} [z(s) \cdot p(s)] ds \\
&= \sum_{j=1}^n \left\{ \frac{\partial}{\partial x_j} \psi(t_1, x^*(t_1)) z_j(t_1) \right. \\
&\quad \left. + \int_r^{t_1} \frac{\partial}{\partial x_j} L(s, x^*(s), u^*(s)) z_j(s) ds \right\},
\end{aligned}$$

which implies that

$$\frac{\partial}{\partial r} J(r, x^*(r); u^*) = -L(r, x^*(r), u^*(r)) - f(r, x^*(r), u^*(r)) \cdot p(r).$$

□

Chapter 4

Singular Stochastic Control

This chapter delves into singular stochastic control problems, which are unique in their allowance for control actions that can cause immediate and discontinuous changes in the state of a system. These types of problems are common in fields like finance, where optimal trading strategies may require sudden large trades, in manufacturing for switching machinery on and off, and in energy management for resource extraction. The chapter starts by introducing the concept of singular stochastic control and discussing its necessity in scenarios where continuous control is not viable or optimal. This is followed by a formalization of the problem, where the state dynamics are described using stochastic differential equations (SDEs) that incorporate control variables capable of inducing jumps. This framework sets the foundation for deriving optimality conditions for these problems. A significant portion of the chapter is devoted to establishing and proving necessary conditions for optimality. This is achieved through the derivation of the Hamilton-Jacobi-Bellman (HJB) equation and the adaptation of verification theorems to the singular control context. These theoretical tools are crucial for determining the optimal control strategies in systems subject to abrupt state changes. The chapter also emphasizes the role of viscosity solutions in dealing with singular stochastic control problems.

4.1 Singular Stochastic Control

We consider an infinite horizon problem where $O \subset \mathbb{R}^n$, $U \subset \mathbb{R}^n$ a closed cone, functions $\hat{f}, \hat{c} \in C^1(\mathbb{R}^n)$ with bounded first order partial derivatives and $\hat{c}, \hat{L} \in C(\mathbb{R}^n)$ such that $f(x, v) = v + \hat{f}(x)$, $\sigma(x, v) = \hat{\sigma}(x)$, $L(x, v) = \hat{L}(x) + \hat{c}(v)$ for all $x \in \mathbb{R}^n, v \in U$, and $\hat{c}$ homogeneous of degree one. Furthermore, we set null boundary condition $g \equiv 0$ and non-negative costs. If we then define

$$\mathcal{H}(t, x, p, A) = \sup_{v \in U} \left[-f(t, x, v) \cdot p - \frac{1}{2} \text{tr}(a(t, x, v)A) - L(t, x, v) \right]. \quad (4.1)$$

Let $\nu = (\Omega, \{\mathcal{F}_s\}, P, w)$ be a reference system, with $\mathcal{F}_s$ right continuous. We want to define the motion of a state variable through a Stochastic Differential Equation. Let us define the auxiliary functions

$$\hat{u}(s) = \begin{cases} |u(s)|^{-1}u(s) & \text{if } u(s) \neq 0 \\ 0 & \text{if } u(s) = 0, \end{cases} \quad (4.2)$$

and

$$\xi(t) = \int_0^t |u(s)| ds. \quad (4.3)$$

We thus define the SDE

$$dx(s) = \hat{f}(x(s))ds + \hat{\sigma}(x(s))dw(s) + \hat{u}(s)d\xi(s), \quad s > 0. \quad (4.4)$$

What do we consider as a control variable in this context? In classical stochastic optimal control problem the SDE is

$$dx(s) = f(x(s), u(s))ds + \sigma(x(s), u(s))dw(s),$$

where both f and σ are possibly time-dependent, and u is the control variable. We define the control variable for (4.4) assign

$$z(t) = \int_{[0,t)} \hat{u}(s) d\xi(s). \quad (4.5)$$

Since we aim at more general $z(\cdot)$ control functions, that may fail to be absolutely continuous, we impose them to be of bounded variation on every interval $[0, t)$, thus obtaining an almost always differentiable functions. If so, given $\mu(\cdot)$ the total variation of $z(\cdot)$, we get

$$\xi(t) = \int_{[0,t)} d\mu(s)$$

which is non-decreasing, left-continuous and $\xi(0) = 0$. Besides, Radon-Nikodym Theorem implies the existence of a function $\hat{u}(s)$ such that (4.5) holds. Under this construction $\mathcal{F}_s$ -measurability of $z(\cdot)$ is passed to $\xi(\cdot)$ and $\hat{u}(\cdot)$, thus we always assume it. Moreover, we assume that $\hat{u}(s) \in U$ for μ -almost all $s \geq 0$ and that each moment of $z(\cdot)$ is finite, that is $E|z(t)|^m < +\infty$ for all $m \in \mathbb{N} \setminus \{0\}$. The existence of a unique solution to (4.4) is proven by Picard iteration, although such a $x(\cdot)$ is not necessarily continuous.

We now want to maximize

$$J(x; \xi, \hat{u}) = E_x \int_{[0,\tau)} e^{-\beta s} \left[\hat{L}(x(s))ds + \hat{c}(\hat{u}(s))d\xi(s) \right], \quad (4.6)$$

over all controls $(\xi(\cdot), \hat{u}(\cdot)) \in \mathcal{A}_\nu$, where τ is the exit time of $x(s)$ from $\overline{O}$. Finally, to avoid the possibility of J being $+\infty$ we impose

$$E_x \int_{[0,\tau)} e^{-\beta s} |L(x(s), u(s))| ds < +\infty. \quad (4.7)$$

Thus, the value functions are

$$V_\nu(x) = \inf_{\mathcal{A}_\nu} J(x; \xi, \hat{u}), \quad (4.8)$$

and

$$V(x) = V_{PM}(x) = \inf_{\nu} V_\nu. \quad (4.9)$$

We would now be tempted to search for solutions of

$$-\frac{\partial V}{\partial t} + \mathcal{H}(t, x, D_x V, D_x^2 V) = 0, \quad (t, x) \in Q,$$

but in this context it may be the case that $\mathcal{H}(p) = +\infty$, indeed if

$$H(p) = \sup_{v \in \hat{K}} -p \cdot v - \hat{c}(v), \quad (4.10)$$

where $\hat{K}$ as the unitary elements of U , is strictly positive value then because of homogeneity and U being a cone then

$$\hat{\mathcal{H}}(p) = \sup_{v \in U} \{-p \cdot v - \hat{c}(v)\} \quad (4.11)$$

is $+\infty$ there, thus observing that

$$\mathcal{H}(x, p, A) = -\frac{1}{2} \text{tr}(\hat{a}(x)A) - \hat{f}(x) \cdot p - \hat{L}(x) + \hat{\mathcal{H}}(p),$$

we get that $\mathcal{H}$ is $+\infty$. Nevertheless, we expect

$$H(DV(x)) \leq 0, \quad (4.12)$$

and

$$\mathcal{L}V(x) = \beta V(x) - \frac{1}{2} \text{tr}(\hat{a}(x)D^2V(x)) - \hat{f}(x) \cdot DV(x) \leq \hat{L}(x), \quad x \in O. \quad (4.13)$$

But if $H(DV(x)) < 0$ then in a neighborhood of x the optimal control is zero, thus we have

$$\mathcal{L}V(x) = \hat{L}(x).$$

In a more compact notation

$$\max \left\{ \mathcal{L}V(x) - \hat{L}(x), H(DV(x)) \right\} = 0, \quad x \in O. \quad (4.14)$$

4.1.1 Verification Theorem

The definition of a classical solution of (4.14) is the following.

Definition 4.1.1. Let $W \in C_p(\overline{O}) \cap C^1(\overline{O})$ with $DW \in L_{loc}^{1,\infty}(O; \mathbb{R}^n)$ and define $\mathcal{P} = \{x \in \mathbb{R}^n : H(DW(x)) < 0\}$. We say that W is classical solution to (4.14) if $W \in C^2(\mathcal{P})$ and $\mathcal{L}W(x) = \hat{L}(x)$ for all $x \in \mathcal{P}$, while $H(DW(x)) \leq 0$ for all $x \in \overline{O}$, and $\mathcal{L}W(x) \leq \hat{L}(x)$ almost everywhere in $\mathbb{R}^n$.

A verification theorem holds in this context. This result mimics one for classical stochastic optimal control problems, that we have not proved in chapter II [cite IV 5.1].

Theorem 4.1.1. Let O be convex and W solution to (4.14) with boundary condition $V(x) = 0$ for $x \in \partial O$. Then for all $x \in \overline{O}$:

(1.) for all $(\xi(\cdot), \hat{u}(\cdot)) \in \hat{\mathcal{A}}_\nu$ such that

$$\liminf_{t \rightarrow +\infty} e^{-\beta t} E_x [W(x(t)) \chi_{\tau=+\infty}] = 0, \quad (4.15)$$

then $W(x) \leq J(x; \xi, \hat{u})$

(2.) if $W \geq 0$ and there exists $(\xi^*(\cdot), u^*(\cdot)) \in \hat{\mathcal{A}}_\nu$ such that with probability one:

(2.a.) $x^*(t) \in \mathcal{P}$ for almost every $t \leq \tau$,

(2.b.) for all $t \leq \tau$

$$\int_{[0,t)} [u^*(s) \cdot DW(x^*(s)) + \hat{c}(u^*(s))] d\xi^*(s) = 0,$$

(2.c.) for all $t \leq \tau$

$$W(x^*(t)) - W(x^*(t^+)) = \hat{c}(u^*(t)) [\xi^*(t^+) - \xi(t)],$$

(2.d.) and

$$\lim_{t \rightarrow +\infty} E_x [e^{-\beta(t \wedge \tau)} W(x^*(t \wedge \tau)) \chi_{\tau=+\infty}] = 0.$$

Then $J(x; \xi^*, u^*) = W(x)$.

Proof. (1.) Let us extend W to $\mathbb{R}^n$ by setting $W \equiv 0$ in $\mathbb{R}^n \setminus \overline{O}$. We identify this extension with W , noticing that it is continuous since W is null on ∂O . We will show an upper bound for W given by a sort-of-mollified version of J , which will give the thesis at the limit. Let us consider a family of mollifiers $\{\varphi_m\}$ centered in the origin. We consider W and $\hat{L}$ mollified by φ_m , that is

$$W_m = W * \varphi_m, \quad \hat{L}_m = \hat{L} * \varphi_m. \quad (4.16)$$

By classic results we have local uniform convergence of W_m , $\hat{L}_m$, and DW_m to W , $\hat{L}$, and DW respectively. Let us now consider

$$O_{N,m} = \left\{ x \in O \mid \text{dist}(x, \partial O) > \frac{1}{m} \right\} \cap B_N(0). \quad (4.17)$$

Now, given the continuity of $\mathcal{L}$ and local boundedness of D^2W we get

$$\lim_{m \rightarrow +\infty} \sup_{x \in \bar{O}_{N,m}} |(\mathcal{L}W_m)(x) - (\mathcal{L}W * \varphi_m)(x)| = 0,$$

while we get

$$\limsup_{m \rightarrow +\infty} K_{N,m} = \limsup_{m \rightarrow +\infty} \sup_{x \in \bar{O}_{N,m}} \left\{ \mathcal{L}W_m(x) - \hat{L}_m(x) \right\} \leq 0,$$

since $\mathcal{L}W \leq \hat{L}$ almost everywhere in O . Hence, since

$$-\hat{v} \cdot DW_m(x) - \hat{c}(\hat{v}) = -[(\hat{v} \cdot DW + \hat{c}(\hat{v})) * \varphi_m](x) \leq 0, \quad x \in \bar{O}_m,$$

we have

$$\max \left\{ \mathcal{L}W_m(x) - \hat{L}_m(x), H(DW_m(x)) \right\} \leq K_{N,m} \vee 0. \quad (4.18)$$

Now for $x \in O_{N,m}$ and $(\xi, \hat{u}) \in \hat{\mathcal{A}}_\nu$ we apply Itô's rule for semimartingales [7] to $e^{-\beta t} W_m(x(t))$ and get

$$\begin{aligned} W_m(x) &= E_x e^{-\beta(t \wedge \tau_m)} W_m(x(t \wedge \tau_m)) + E_x \int_0^{t \wedge \tau_m} e^{-\beta s} \mathcal{L}W_m(x(s)) ds \\ &\quad + E_x \int_0^{t \wedge \tau_m} -e^{-\beta s} u(s) \cdot DW_m(x(s)) d\xi^c(s) \\ &\quad + E_x \sum_{0 \leq s < t \wedge \tau_m} e^{-\beta s} [W_m(x(s)) - W_m(x(s^+))], \end{aligned}$$

for a given $N > 0$ and where τ_m is the time from $\bar{O}_{N,m}$ and $\xi^c(t)$ is $\xi(t)$ minus all the jumps until then:

$$\xi^c(t) = \xi(t) - \sum_{0 \leq s < t} \xi(s^+) - \xi(s).$$

If we let $m \rightarrow +\infty$ then

$$\begin{aligned} W(x) &\leq E_x e^{-\beta(t \wedge \omega_N)} W(x(t \wedge \omega_N)) + E_x \int_0^{t \wedge \omega_N} e^{-\beta s} \hat{c}(\hat{u}(s)) d\xi^c(s) \\ &\quad + E_x \sum_{0 \leq s < t \wedge \omega_N} e^{-\beta s} \hat{c}(\hat{u}(s)) (\xi(s^+) - \xi(s)), \end{aligned}$$

but if $N \rightarrow +\infty$ then

$$W(x) \leq E_x e^{-\beta(t \wedge \tau)} W(x(t \wedge \tau)) + E_x \int_0^{t \wedge \tau} e^{-\beta s} (\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s)),$$

which yields

$$W(x) \leq E_x e^{-\beta(t \wedge \tau)} W(x(\tau)) \chi_{\tau < \infty} + E_x \int_0^\tau e^{-\beta s} (\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) dx(s)).$$

If we prove that for $\tau < \infty$ we have

$$W(x(\tau)) \leq \hat{c}(\hat{u}(\tau)) [\xi(\tau^+) - \xi(\tau)], \quad (4.19)$$

then the thesis holds true. The boundary condition implies (4.19) for $x(\tau) \in \partial O$. If $x(\tau) \in O$ then there exists x^* between $x(\tau)$ and $x(\tau^+)$ on the boundary of O . Thus,

$$-\hat{u}(\tau) \cdot DW(x_\lambda) - \hat{c}(\hat{u}(\tau)) \leq H(DW(x_\lambda)) \leq 0,$$

for $x_\lambda = x(\tau) + \lambda(x^* - x(\tau))$ for all $\lambda \in [0, 1]$. Since

$$x^* - x(\tau) = \gamma \hat{u}(\tau) (\xi(\tau^+) - \xi(\tau)),$$

then we get

$$\begin{aligned} W(x(\tau)) &= W(x^*) - \int_0^1 DW(x_\lambda) d\lambda \cdot (x^* - x(\tau)) \\ &= -\gamma \int_0^1 \hat{u}(\tau) \cdot DW(x_\lambda) d\lambda [\xi(\tau^+) - \xi(\tau)] \leq \hat{c}(\hat{u}(\tau)) [\xi(\tau^+) - \xi(\tau)]. \end{aligned}$$

(2.) let us suppose that for some $s \geq 0$ we have $x^*(s) \in \mathcal{P}$. Since $\mathcal{P}$ is open

$$\mathcal{L}W_m(x^*(s)) \xrightarrow{m \rightarrow +\infty} \mathcal{L}W(x^*(s)) = \hat{L}(x^{ast}(s)).$$

By (2.a.) and the dominated convergence theorem we have

$$\lim_{m \rightarrow +\infty} \int_0^{t \wedge \theta_m} e^{-\beta s} \mathcal{L}W_m(x^*(s)) ds = E_x \int_0^{t \wedge \theta_N} e^{-\beta s} \hat{L}(x^*(s)) ds. \quad (4.20)$$

If we now impose the optimal control into (??) we get

$$\begin{aligned} W_m(x) &= E_x e^{-\beta(t \wedge \tau_m)} W_m(x^*(t \wedge \tau_m)) \\ &\quad + E_x \int_0^{t \wedge \tau_m} e^{-\beta s} \mathcal{L}W_m(x^*(s)) ds \\ &\quad + E_x \int_0^{t \wedge \tau_m} e^{-\beta s} [-u^*(s) \cdot DW_m(x^*(s))] d\xi^{*,c}(s) \\ &\quad + E_x \sum_{0 \leq s < t \wedge \tau_m} e^{-\beta s} [W_m(x^{ast}(s)) - W_m(x^*(s^+))], \end{aligned} \quad (4.21)$$

thus letting $m, N, t \rightarrow +\infty$ and recalling (2.b.), (2.c.), and (2.d.) we get the thesis. $\square$

4.2 Viscosity Solution

Consistently with what has been done in previous chapters we now prove the value function arising from the Singular stochastic problem is indeed a solution (viscous) of the dynamic programming equation. We have already defined the dynamic programming equation in this context, that is equation (4.14); the value function V is assumed to be in $C_p(\overline{O})$ and satisfy the dynamic programming principle, that is

$$V(x) = \inf_{\nu} \inf_{\mathcal{A}_{\nu}} \left\{ E_x e^{-\beta(\tau \wedge \theta)} V(x(\tau \wedge \theta)) + E_x \int_{[0, \tau \wedge \theta]} e^{-\beta s} \left[\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s) \right] \right\}, \quad (4.22)$$

where τ is the exit time of $x(\cdot)$ from $\overline{O}$ and (4.22) holds for all $x \in O$ and all stopping times θ . Formally we can also define the value function as

$$V(x) = \inf_{\nu} \inf_{\mathcal{A}_{\nu}} J(x; \xi, u), \quad (4.23)$$

recalling what we did in previous chapters.

Theorem 4.2.1. *Let $V \in C_p(\overline{O})$ such that (4.22) holds $\forall x \in O$ and all stopping times θ , then V is a viscosity solution of (4.14).*

Proof. (Subsolution.) Shall we take a $w \in C^2(O) \cap C_p(\overline{O})$ and $\bar{x} \in O$ such that

$$\bar{x} = \arg \max_{\overline{O}} V - w$$

and show that

$$\max \left\{ \mathcal{L}w(\bar{x}) - \hat{L}(\bar{x}), H(Dw(\bar{x})) \right\} \leq 0. \quad (4.24)$$

From (??) by taking $\theta = h > 0$ we get

$$w(\bar{x}) \leq E_{\bar{x}} \int_{[0, \tau \wedge h]} e^{-\beta s} \left[\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s) \right] + E_{\bar{x}} e^{-\beta(\tau \wedge h)} w(x(\tau \wedge h)). \quad (4.25)$$

If we set $\delta > 0$, $\xi = \delta \chi_{\mathbb{R}_+}$, and $\hat{u}(s) \equiv \hat{v} \in U$ we get

$$x(0^+) = \lim_{t \rightarrow 0^+} x(t) = x(0) + \hat{u}(0) \left(\lim_{t \rightarrow 0^+} \xi(t) - \xi(0) \right) = \bar{x} + \hat{v}\delta,$$

then if we let $h \rightarrow 0^+$ in (4.25) we get

$$w(\bar{x}) \leq \delta \hat{c}(\hat{v}) + w(\bar{x} + \delta \hat{v}), \quad (4.26)$$

which implies

$$-Dw(\bar{x}) - \hat{c}(\hat{v}) = \lim_{\delta \rightarrow 0} \frac{w(\bar{x} + \delta \hat{v}) - w(\bar{x})}{\delta} - \hat{c}(\hat{v}) \leq 0, \quad (4.27)$$

for all $\hat{v} \in U$, thus

$$H(Dw(\bar{x})) \leq 0.$$

If we instead impose $\xi = 0$ in (4.25) we get

$$w(\bar{x}) \leq E_{\bar{x}} \int_{[0, \tau \wedge h]} e^{-\beta s} \hat{L}(x(s)) ds + E_{\bar{x}} e^{-\beta(\tau \wedge h)} w(x(\tau \wedge h)), \quad (4.28)$$

if we then apply Ito's formula to $e^{-\beta s} w(x(s))$ we get

$$E_{\bar{x}} e^{-\beta(\tau \wedge h)} w(x(\tau \wedge h)) = -E_{\bar{x}} \int_{[0, \tau \wedge h]} e^{-\beta s} \mathcal{L}w(x(s)) ds + w(\bar{x}), \quad (4.29)$$

then combining (4.28) together with (4.29) we get

$$E_{\bar{x}} \int_{[0, \tau \wedge h]} e^{-\beta s} \mathcal{L}(x(s)) - e^{-\beta s} \hat{L}w(x(s)) ds \leq 0, \quad (4.30)$$

thus letting $h \rightarrow 0^+$

$$\mathcal{L}w(x(s)) - \hat{L}(x(s)) \leq 0. \quad (4.31)$$

(Supersolution.) Let w as before and $\bar{x} \in O$ such that $V - w$ is minimized with $(V - w)(\bar{x}) = 0$, let us show

$$\max \left\{ \mathcal{L}w(\bar{x}) - \hat{L}(\bar{x}), H(Dw(\bar{x})) \right\} \geq 0. \quad (4.32)$$

Let us suppose the contrary, that is the existence of $\delta, \gamma > 0$ such that

$$\max \left\{ \mathcal{L}w(\bar{x}) - \hat{L}(\bar{x}), H(Dw(\bar{x})) \right\} \leq -\gamma, \quad \forall x \in B_{\gamma}(\bar{x}). \quad (4.33)$$

Let $(\xi(\cdot), \hat{u}(\cdot)) \in \hat{\mathcal{A}}_{\nu}$ and θ the exit time from the ball of $x(\cdot)$, thus $\theta < \tau$. Since $w \in C^2$ then $e^{-\beta s} w(x(s))$ is a semi-martingale, and we can apply Ito's formula to semi-martingales

$$\begin{aligned} w(\bar{x}) &= E_{\bar{x}} e^{-\beta \theta} w(x(\theta)) + E_{\bar{x}} \int_0^{\theta} e^{-\beta s} \mathcal{L}w(x(s)) ds \\ &\quad + E_{\bar{x}} \int_0^{\theta} e^{-\beta s} [-\hat{u}(s) \cdot Dw(x(s))] d\xi^c(s) \\ &\quad + E_{\bar{x}} \sum_{0 \leq s < \theta} e^{-\beta s} [w(x(s)) - w(x(s^+))], \end{aligned} \quad (4.34)$$

with $\xi^c(t) = \xi(t) - \sum_{0 \leq s < t} \xi(s^+) \xi(s)$. Clearly the absurd condition implies

$$\mathcal{L}w(x(s)) \leq \hat{L}(x(s)) - \gamma, \quad H(Dw(x(s))) \leq -\gamma, \quad (4.35)$$

while we get

$$-\hat{u}(s) \cdot Dw(x(s)) \leq \hat{c}(\hat{u}(s)) - \gamma |\hat{u}(s)| \quad (4.36)$$

by using the latter together with (4.10). Furthermore, by the fundamental theorem of calculus (FTC) if we set $x_\rho = \rho x(s^+) + (1 - \rho)x(s)$ for all $\rho \in [0, 1]$ we have

$$\begin{aligned} w(x(s)) - w(x(s^+)) &= \int_0^1 -(x(s^+) - x(s)) \cdot Dw(\rho x(s^+) + (1 - \rho)x(s)) d\rho \\ &= \int_0^1 -(x(s^+) - x(s)) \cdot Dw(x_\rho) d\rho \\ &\leq [\hat{c}(\hat{u}(s) - \gamma|\hat{u}(s)|)] [\xi(s^+) - \xi(s)]. \end{aligned}$$

Thus, if we combine the latter with (4.36) and (4.35) into (4.34) we get

$$\begin{aligned} w(\bar{x}) &\leq E_{\bar{x}} e^{-\beta\theta} w(x(\theta)) \\ &\quad + E_{\bar{x}} \int_{[0, \theta)} e^{-\beta s} \left[\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s) \right] \\ &\quad - \gamma E_{\bar{x}} \int_{[0, \theta)} e^{-\beta s} [ds + |\hat{u}(s)| d\xi(s)]. \end{aligned} \tag{4.37}$$

Moreover, there exists $\lambda \in [0, 1]$ random variable such that

$$x_\lambda = x(\theta) + \lambda \hat{u}(\theta) [\xi(\theta^+) - \xi(\theta)] \in \partial B_\delta(\bar{x}),$$

since $x(\theta^+) \notin B_\delta(\bar{x})$ and $x(\theta^+) = x(\theta) + \hat{u}(\theta) [\xi(\theta^+) - \xi(\theta)]$. Rewriting via the FTC once again we get

$$w(x(\theta)) \leq w(x_\lambda) + \lambda [\hat{c}(\hat{u}(\theta)) - \gamma|\hat{u}(\theta)|] [\xi(\theta^+) - \xi(\theta)].$$

Thus, we have that

$$w(x(\theta)) \leq V(x(\theta^+)) + [\hat{c}(\hat{u}(\theta)) - \lambda\gamma|\hat{u}(\theta)|] [\xi(\theta^+) - \xi(\theta)],$$

this comes from the obvious inequality

$$V(x) - V(x + hv) \leq h\hat{c}(v), \quad \forall v \in U$$

and sufficiently small $h > 0$, when $x(\theta^+) \in \bar{O}$, and from the same reasoning used while proving (4.19), that is via convexity of O and FTC if $x(\theta^+) \notin \bar{O}$. Recalling (4.37)

$$\begin{aligned} w(\bar{x}) &\leq E_{\bar{x}} e^{-\beta\theta} V(x(\theta)) \\ &\quad + E_{\bar{x}} \int_{[0, \theta]} e^{-\beta s} \left[\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s) \right] \\ &\quad - \gamma E_{\bar{x}} \int_{[0, \theta)} e^{-\beta s} [ds + |\hat{u}(s)| d\xi(s)] \\ &\quad - E_{\bar{x}} e^{-\beta\theta} \lambda\gamma|\hat{u}(\theta)| [\xi(\theta^+) - \xi(\theta)]. \end{aligned} \tag{4.38}$$

If we prove the existence of $\alpha_0 > 0$ such that

$$E_{\bar{x}} \int_{[0, \theta)} e^{-\beta s} [ds + |\hat{u}(s)| d\xi(s)] + E_{\bar{x}} e^{-\beta \theta} \lambda |\hat{u}(\theta)| [\xi(\theta^+) - \xi(\theta)] \leq \alpha_0, \quad (4.39)$$

for all controls over all systems, then (4.38) implies

$$\begin{aligned} V(\bar{x}) &\leq E_{\bar{x}} e^{-\beta \theta} V(x(\theta)) \\ &\quad + E_{\bar{x}} \int_{[0, \theta]} e^{-\beta s} [\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s)] \\ &\quad - \gamma E_{\bar{x}} \int_{[0, \theta)} e^{-\beta s} [ds + |\hat{u}(s)| d\xi(s)] \\ &\quad - \gamma \alpha_0, \end{aligned}$$

which contradicts

$$V(x) = \inf_{\nu} \inf_{\mathcal{A}_{\nu}} \left\{ E_x e^{-\beta(\theta)} V(x(\theta^+)) + E_x \int_{[0, \theta]} e^{-\beta s} [\hat{L}(x(s)) ds + \hat{c}(\hat{u}(s)) d\xi(s)] \right\},$$

which is (4.22) with stopping time $\theta + h$ and $h \rightarrow 0$. Equation (4.39) holds true since the lhs can be seen as the payoff of a singular stochastic control problem $\tilde{J}(x; \tilde{\xi}, \hat{u})$ where $\tilde{\xi} = \xi \chi_{t \leq \theta} + (\xi(\theta) + \lambda(\xi(\theta^+) - \xi(\theta))) \chi_{t > \theta}$, then if we define

$$W(x) = C(\delta^2 - |x - \bar{x}|^2), \quad x \in B_{\delta}(\bar{x}),$$

we get

$$\tilde{V}(\bar{x}) \geq \tilde{J}(\bar{x}; \tilde{\xi}, \hat{u}) \geq W(\bar{x}) = \left(\sup_{|x - \bar{x}| \leq \delta} \left\{ \beta \delta^2 + \text{tr}(\hat{a}(x)) + 2 \left| \hat{f}(x) \cdot (x - \bar{x}) \right| \right\} \wedge \frac{1}{2\delta} \right) \delta^2.$$

□

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